

FUZZY SET& FUZZY LOGIC, AND ITS APPLICATIONS

Project report submitted to

Mahatma Gandhi University, Kottayam

*For the partial fulfillment of the requirements for the award of degree of
Bachelor of Science in Mathematics.*

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2021-2024



CERTIFICATE

This is to certify that the dissertation with the title “**Fuzzy sets & Fuzzy logic, And its Application**” is a bonafide record of the work done by **Abhinav S, Ajith K C, Akash K S & Arunkumar S** under the supervision and guidance of **Prof. Prathesh Mathew**, Department of Mathematics, Nirmala College Muvattupuzha in partial fulfillment of the requirements for the award of **Bachelor’s Degree in Mathematics** of **Mahatma Gandhi University, Kottayam**.

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DECLARATION

We hereby declare that this project report titled **“Fuzzy sets & Fuzzy logic, And its Application”** in partial fulfillment of the requirements for the award of **Bachelor’s Degree in Mathematics of Mahatma Gandhi University, Kottayam**, is our dissertation work. The contents of the study, in full or parts have not been submitted to any other institution or university for the award of degree or diploma.

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ACKNOWLEDGEMENT

We take this opportunity to express our sincere thanks to all who have contributed towards the successful completion of this project study First and foremost, we bow before **The Almighty**.

With great pleasure we express our sincere gratitude to our guide **Prof. Pratheesh Mathew** Asst. Professor, Department of Mathematics, Nirmala College, Muvattupuzha for the completion of this work successfully.

We further express our sense of gratitude to Dr. Thomas K V, Principal, **Prof. Pratheesh Mathew**, Head of the Department of Mathematics, Nirmala College, Muvattupuzha and other teachers of the department for the encouragement, suggestions and assistance in taking up this dissertation.

We are most grateful to the library staff for the help extended in accomplishing this work. We finally thank our parents, friends and all our well-wishers who had supported us directly or indirectly, during the project.

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INTRODUCTION

Fuzzy set theory was proposed by Zadeh in 1965 as an extension of the classical notion of a set (Zadeh, 1965). With the proposed methodology, Zadeh introduced a mathematical method with which decision-making using fuzzy descriptions of some information becomes possible.

This project is a natural outgrowth of Fuzzy Sets, Uncertainty, and Information and It reflects the tremendous advances that have taken place in the areas of fuzzy sets. The capability of fuzzy sets to express gradual transitions from membership to non - membership and vice versa has a broad utility. It provides us not only with a meaningful and powerful representation of measurement uncertainties but also with a meaningful representation of vague concepts expressed in natural language. Also, we introduce the basic concept of fuzzy logic, which is an approach to computing based on “degrees of truth” rather than the usual “true or false”. The idea of fuzzy logic was first advanced by Lotfi Zadeh of the University of California.

In this book, we present an introduction to the major developments of fuzzy sets and fuzzy logic as well as to some of the most successful applications of the theory. To enhance comprehension of the topic, we provide numerous examples.

The project "FUZZY SET & FUZZY LOGIC, AND ITS APPLICATIONS" comprises five modules. The initial module covers preliminaries. Following that, the second module addresses fuzzy operations and fuzzy arithmetic. The third module delves into fuzzy relations and fuzzy graphs. Subsequently, the fourth module introduces the concept of fuzzy logic, while the final module explores various applications of fuzzy sets.

MODULE 1: PRELIMINARIES

1.1: Crisp Sets

In this chapter, we refresh the basic concepts of crisp sets and introduce the notation and terminology specific to fuzzy sets.

- Sets are usually denoted by upper case letters and their members by lower case letters.
- The letter U denotes the universal set.
- The set which contains no members is called an empty set and is denoted by φ .
- To indicate that an object x is a member of a set A , we write $x \in A$.

Conversely, $x \notin A$ denotes that x is not a member of A .

Methods of Defining a Set

1. List Method (Roaster method)
2. Rule Method (Set Builder Method)
3. Characteristic Function

(i) List Method (Roaster Method)

In this method, a set is defined by naming all its members. It can be used only for finite sets.

If a set A has elements $a_1, a_2, \dots, a_n$, then it is usually denoted as:

$$A = \{a_1, a_2, \dots, a_n\}$$

(ii) Rule Method (Set Builder Method)

In this method the set is aligned by a property satisfied by its members. A common notation expressing the method is written as

$$A = \{x \mid p(x)\}$$

Where $p(x)$ is a proposition of the form " x has the property $P(x)$ ".

- If every member of a set A is also a member of set B then A is called a subset of B . It is denoted as $A \subset B$ where A is called subset of B and B is called superset of A .

- If two sets are equal then it is called equal sets. It is denoted by $A=B$
- If the elements of the set A and the set B are not equal then it is called unequal set. It is denoted by $A \neq B$

(iii) Characteristic Function

A set is defined by a function, usually called a characteristic function, that declares which elements of X are members of the set and which are not. Set A is defined by its characteristic function, χ_A , as follows:

$$\chi_A(x) = \begin{cases} 1 & \text{for } x \in A \\ 0 & \text{for } x \notin A \end{cases}$$

Operations on sets

1.Union:

The union of set A and B is the set that contains all elements that belong to either set A or set B or both sets. It is denoted as $A \cup B$. Then

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

2.Intersection:

The intersection of sets A and B is the set containing all the elements belonging to both set A and set B. It is denoted by $A \cap B$. Thus,

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

3.Difference:

The difference of a set A with respect to B is the set containing all the members of B that are not also members of A. This can be written $B - A$. Thus,

$$B - A = \{x \mid x \in B \text{ and } x \notin A\}.$$

4.Complement:

The complement of a set A is the set consisting of all the elements of the universal set except A.

It is denoted by A' . Thus, $A' = \{x \mid x \in U \text{ and } x \notin A\}$

FUNDAMENTAL PROPERTIES OF CRISP SET OPERATIONS

Involution	$\overline{\overline{A}} = A$
Commutativity	$A \cup B = B \cup A$ $A \cap B = B \cap A$
Associativity	$(A \cup B) \cup C = A \cup (B \cup C)$ $(A \cap B) \cap C = A \cap (B \cap C)$
Distributivity	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
Idempotence	$A \cup A = A$ $A \cap A = A$
Absorption	$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$
Absorption by X and $\emptyset$	$A \cap X = A$ $A \cup \emptyset = A$
Identity	$A \cup \emptyset = A$ $A \cap X = A$
Law of contradiction	$A \cap \emptyset = \emptyset$ $A \cup X = X$
Law of excluded middle	$A \cup A = X$ $A \cap A = A$
De Morgan's laws	$(A \cup B)' = A' \cap B'$ $(A \cap B)' = A' \cup B'$

1.2 Fundamentals of Fuzzy Sets

The characteristic function of a crisp set assigns each element a value of either 1 or 0, discriminating between members and non-members of the crisp set under consideration. But this function can be generalized to the range $[0, 1]$.

Thus, the membership function maps each element of the universal set X to a membership grade between 0 and 1. This grade represents the degree of belongingness of the element to the fuzzy set.

The most commonly used range of values for membership functions is the unit interval $[0, 1]$.

The common notations to denote membership functions are:

(i) $\mu_A: X \rightarrow [0, 1]$

(ii) $A: X \rightarrow [0, 1]$

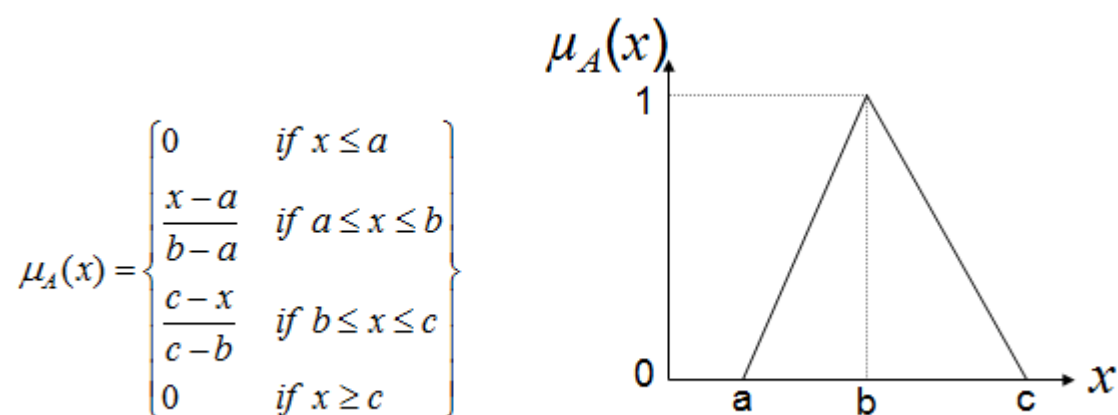
Membership functions can either be chosen by the user arbitrarily, based on the user's experience (MF chosen by two users could be different depending upon their experiences, perspectives, etc.) or be designed using machine learning methods (e.g., artificial neural networks, genetic algorithms, etc.)

There are different shapes of membership functions:

- Triangular,
- Trapezoidal,
- Gaussian etc

Triangular membership function

A triangular membership function is specified by three parameters $\{a, b, c\}$ a , b and c represent the x coordinates of the three vertices of $\mu_A(x)$ in a fuzzy set A (a : lower boundary and c : upper boundary where membership degree is zero, b : the centre where membership degree is 1)



Trapezoid membership function

- A trapezoidal membership function is specified by four parameters $\{a, b, c, d\}$

as follows:

$$\mu_A(x) = \begin{cases} 0 & \text{if } x \leq a \\ \frac{x-a}{b-a} & \text{if } a \leq x \leq b \\ 1 & \text{if } b \leq x \leq c \\ \frac{d-x}{d-c} & \text{if } c \leq x \leq d \\ 0 & \text{if } d \leq x \end{cases}$$

Defining Fuzzy Sets

If X is a collection of objects, then a fuzzy set $\tilde{A}$ in X is a set of ordered pairs:

$$\tilde{A} = \{(x, \mu_A(x)) / x \in X\}$$

where $\mu_A(x)$ is the membership function.

Example: A realtor wants to classify the house he offers to his clients. One indicator of comfort of these houses is the number of bedrooms in it. Let $X = \{1, 2, 3, 4, \dots, 10\}$ be the set of available types of houses described by x = number of bedrooms in a house. Then the fuzzy set "comfortable type of house for a 4-person family" may be described as

$$\tilde{A} = \{(1, 2), (2, 5), (3, 8), (4, 1), (5, 7), (6, 3)\}$$

In the literature one finds different ways of denoting fuzzy sets:

A fuzzy set is denoted by an ordered set of pairs, the first element of which denotes the element and the second the degree of membership.

1.3 Concepts of α -cut and strong α -cut

One of the most important concepts of fuzzy sets is the concept of an α -cut and its variant, a strong α -cut.

(i) α -cut:

Let A be any number Set defined on the universal set X and $\alpha \in [0, 1]$. Then the α -cut ${}^\alpha A$ of the fuzzy Set A is the crisp set that contains all the elements of the universal set X whose membership grades in A are greater than or equal to the specified value of α .

It is mathematically defined by,

$${}^{\alpha}A = \{x \mid A(x) \geq \alpha\}$$

(ii) Strong α -cut:

Let A be any number Set defined on the universal set X and $\alpha \in [0,1]$. The strong α -cut ${}^{\alpha}A$ of the fuzzy Set A is the crisp set that contains all the elements of the universal set X whose membership grades in A are strictly greater than the specified value of α .

It is mathematically defined by,

$${}^{\alpha+}A = \{x \mid A(x) > \alpha\}$$

As an example, consider three fuzzy sets that represent the concepts of a young, middle-aged, and old person. A reasonable expression of these concepts by trapezoidal membership functions A1, A2, and A3.

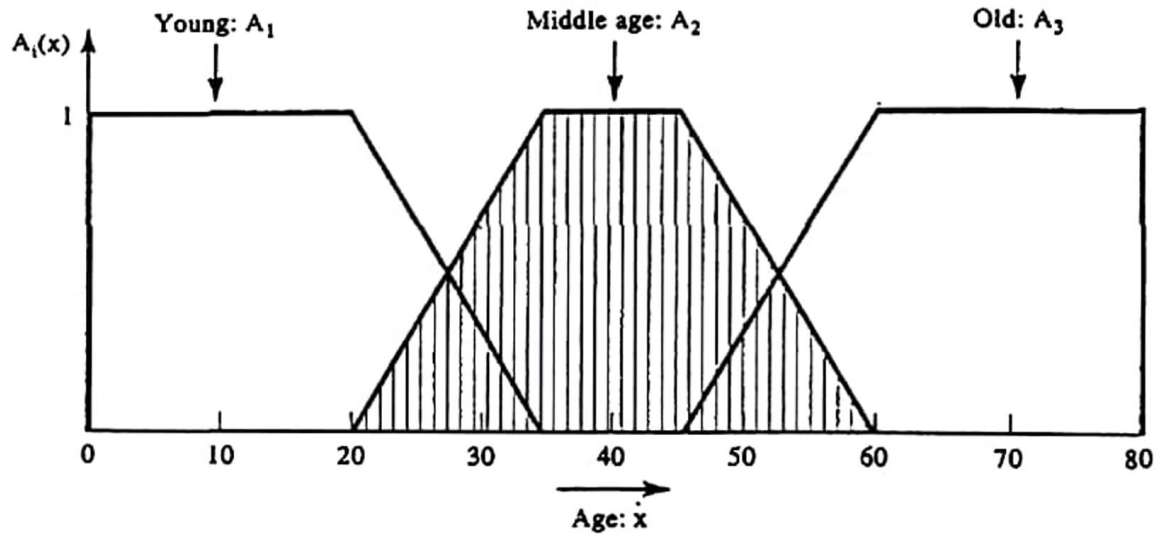
These are defined on the interval $[0, 80]$ as follows,

$$A_1(x) = \begin{cases} 1 & , x \leq 20 \\ (35 - x)/15 & , 20 < x < 35 \\ 0 & , x \geq 35 \end{cases}$$

$$A_2(x) = \begin{cases} 0 & , \text{either } x \leq 20 \text{ or } \geq 60 \\ (x - 20)/15 & , 20 < x < 35 \\ (60 - x)/15 & , 45 < x < 60 \\ 1 & , 35 \leq x \leq 45 \end{cases}$$

$$A_3(x) = \begin{cases} 0 & , x \leq 45 \\ (x - 20)/15 & , 45 < x < 60 \\ 1 & , x \geq 60 \end{cases}$$

A complete characterization of all α -cuts and all strong α -cuts for the fuzzy sets A_1, A_2, A_3 are given in the following figure.



Membership functions representing the concepts of a young, middle-aged, and old person. Shown discrete approximation D_2 of A_2 is defined numerically.

DISCRETE APPROXIMATION OF MEMBERSHIP FUNCTION A_2 BY
FUNCTION D_2 OF THE FORM: $D_2: \{0, 2, 4, \dots, 80\} \rightarrow [0, 1]$

x	$D_2(x)$
$x \notin \{22, 24, \dots, 58\}$	0.00
$x \in \{22, 58\}$	0.13
$x \in \{24, 56\}$	0.27
$x \in \{26, 54\}$	0.40
$x \in \{28, 52\}$	0.53
$x \in \{30, 50\}$	0.67
$x \in \{32, 48\}$	0.80
$x \in \{34, 46\}$	0.93
$x \in \{36, 38, \dots, 44\}$	1.00

$$A_1 = A_2 = A_3 = [0, 80] = X;$$

$$\alpha A_1 = [0, 35 - 15\alpha] \quad \alpha \in (0, 1];$$

$$\alpha A_2 = [15\alpha + 20, 60 - 15\alpha], \quad \alpha A_3 = [15\alpha + 45, 80] \quad \text{for } \alpha \in (0, 1];$$

$$\alpha^+ A_1 = (0, 35 - 15\alpha) \quad \alpha^+ A_2 = (15\alpha + 20, 60 - 15\alpha), \quad \alpha^+ A_3 = (15\alpha + 45, 80) \quad \text{for } \alpha \in [0, 1);$$

$${}^{1+}A_1 = {}^{1+}A_2 = {}^{1+}A_3 = \emptyset.$$

Support of a fuzzy set

The support of a fuzzy set A within a universal set X is the crisp set that contains all the elements in X that have non-zero membership grades in A .

$$S(A) = \{x \mid A(x) > 0\}$$

It is also denoted by $\text{supp}(A)$, $S(A)$, or ${}^{0+}A$.

Height of a fuzzy set

The height $h(A)$ of a fuzzy set A is the longest membership grade obtained by any element in the set.

$$h(A) = \sup A(x)$$

Normalised fuzzy set

A fuzzy set A is called normal when $h(A) = 1$

Note: A fuzzy set A is called sub-normal when $h(A) < 1$

Convex fuzzy set

A fuzzy set is convex if its α -cuts are convex for all $\alpha \in (0, 1)$.

$$\text{i.e., } x_1, x_2 \in {}^{\alpha}A \Rightarrow \lambda x_1 + (1 - \lambda) x_2 \in {}^{\alpha}A$$

$$\forall \lambda \in [0, 1]$$

Theorem 1.1. A fuzzy set A on R is convex if and only if $A(\lambda x_1 + (1 - \lambda) x_2) \geq \min[A(x_1), A(x_2)]$

for all $x_1, x_2 \in R$ and all $\lambda \in [0, 1]$, where $\min$ denotes the minimum operator.

Proof:

(i) Assume that A is convex and let $\alpha = A(x_1) \leq A(x_2)$. Then, $x_1, x_2 \in {}^{\alpha}A$ and, moreover, $\lambda x_1 + (1 - \lambda) x_2 \in {}^{\alpha}A$ for any $\lambda \in [0, 1]$ by the convexity of A . Consequently, $A(\lambda x_1 + (1 - \lambda) x_2) \geq \alpha = A(x_1) = \min[A(x_1), A(x_2)]$.

(ii) Assume that A satisfies (1.13). We need to prove that for any $\alpha \in (0, 1]$, ${}^{\alpha}A$ is convex. Now for any $x_1, x_2 \in {}^{\alpha}A$ (i.e., $A(x_1) \geq \alpha$, $A(x_2) \geq \alpha$), and for any $\lambda \in [0, 1]$, by (1.13)

$$A(\lambda x_1 + (1 - \lambda) x_2) \geq \min[A(x_1), A(x_2)] \geq \min(\alpha, \alpha) = \alpha;$$

i.e., $\lambda x_1 + (1 - \lambda) x_2 \in {}^{\alpha}A$. Therefore, ${}^{\alpha}A$ is convex for any $\alpha \in (0, 1]$. Hence, A is convex.

Fuzzy Number

To qualify as a fuzzy number, a fuzzy set A on $\mathbb{R}$ must possess at least the following properties:

- i. A must be a normal fuzzy set.
- ii. ${}^{\alpha}A$ must be a closed interval for every $\alpha \in (0,1]$
- iii. the support of A , ${}^{0+}A$ must be bounded.

MODULE 2: FUZZY OPERATIONS AND FUZZY ARITHMETIC

2.1 Introduction

The standard fuzzy operations program precisely on the corresponding operations for crisp sets when the range of membership grades is restricted to the set $\{0,1\}$. That is, the standard fuzzy operations are generalizations of the corresponding classical set operations. Fuzzy arithmetic is based on two properties of fuzzy numbers: (1) each fuzzy set, and thus also each fuzzy number, can fully and uniquely be represented by its alpha-cuts and (2) alpha-cuts of each fuzzy number are closed intervals of real numbers for all $\alpha \in (0, 1]$. These properties enable us to define arithmetic operations on fuzzy numbers in terms of arithmetic operations on their α -cuts (i.e., arithmetic operations on closed intervals).

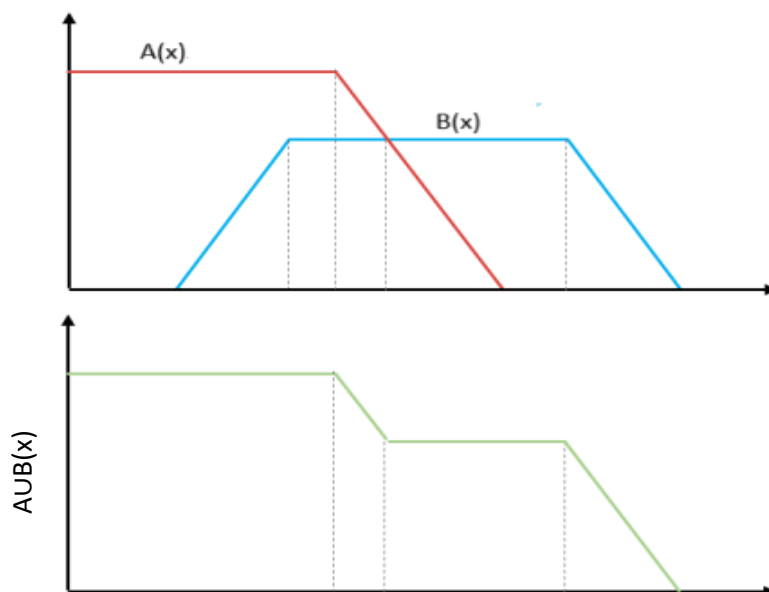
2.2 Operations on Fuzzy Sets

(i) Union:

Let A and B are two fuzzy sets then the union of A and B is defined for all $x \in X$ such that,

$$(A \cup B)(x) = \text{Max} \{A(x), B(x)\}$$

Graphically, we can represent union operations as follows: Red and Blue membership functions represent the fuzzy value for elements in sets A and B, respectively. Wherever these fuzzy functions overlap, we have to consider the point with the maximum membership value.



EXAMPLE:

Let $X = \{a, b, c, d\}$

$A = \{(a,0.2), (b,0.4), (c,0.6), (d,0.9)\}$

$B = \{(a,0.1), (b,0.5), (c,0.7), (d,0.8)\}$

$A \cup B = \{(a,0.2), (b,0.5), (c,0.7), (d,0.9)\}$

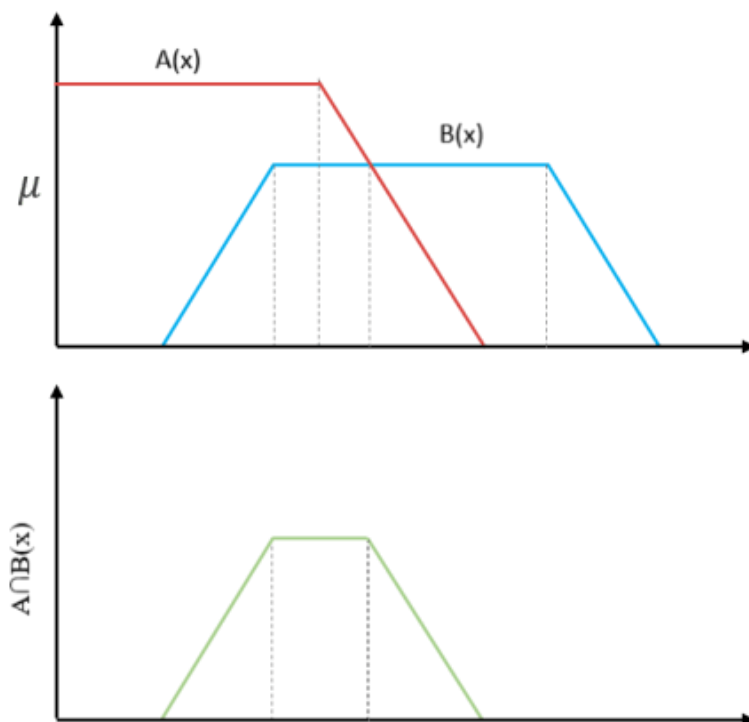
(ii) Intersection:

Given two fuzzy sets, A and B, the standard intersection $A \cap B$ is defined for all $x \in X$ by the equation

$$(A \cap B)(x) = \min [A(x), B(x)]$$

where min denotes the minimum operator.

Graphically, we can represent the intersection operation as follows: Red and blue membership functions represent the fuzzy value for elements in sets A and B, respectively. Wherever these fuzzy functions overlap, we have to consider the point with the minimum membership value.



EXAMPLE:

Let $X = \{a, b, c, d\}$

$A = \{(a, 0.2), (b, 0.4), (c, 0.6), (d, 0.9)\}$

$B = \{(a, 0.1), (b, 0.5), (c, 0.7), (d, 0.8)\}$

$A \cap B = \{(a, 0.1), (b, 0.4), (c, 0.6), (d, 0.8)\}$

(iii) Complement:

The standard complement $\bar{A}$, of a fuzzy set A with respect to the universal set X is defined for all x belongs to X by the equation,

$\bar{A}(x) = 1 - A(x)$ for all x belongs to X .

EXAMPLE:

Let $X = \{a, b, c, d\}$ and let

Fuzzy set $A = \{(a, 0.2), (b, 0.5), (c, 0.3), (d, 0.9)\}$

So, $\bar{A}(a) = 1 - A(a) = 1 - 0.2 = 0.8$

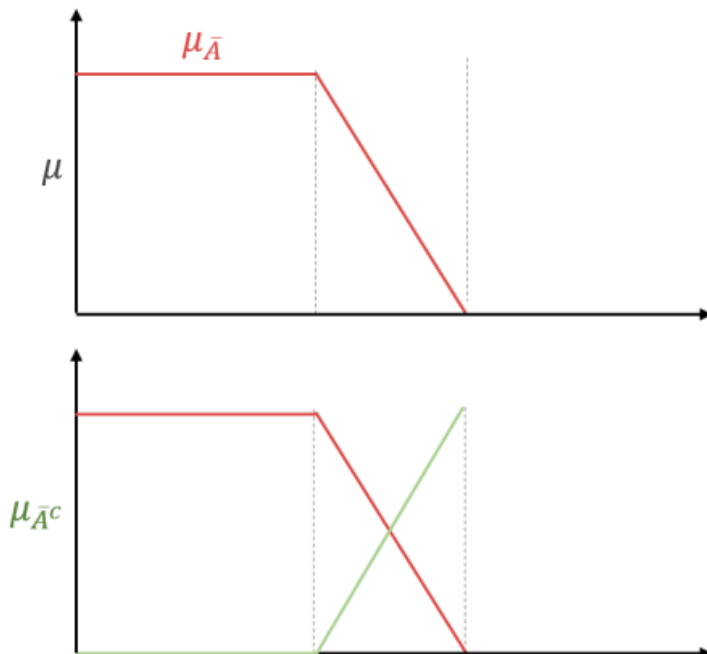
$\bar{A}(b) = 1 - A(b) = 1 - 0.5 = 0.5$

$\bar{A}(c) = 1 - A(c) = 1 - 0.3 = 0.7$

$\bar{A}(d) = 1 - A(d) = 1 - 0.9 = 0.1$

$\bar{A} = \{(a, 0.8), (b, 0.5), (c, 0.7), (d, 0.1)\}$

Graphical representation of fuzzy complement



2.3 Arithmetic operations on intervals

Fuzzy arithmetic is based on two properties of fuzzy numbers: (1) each fuzzy set, and thus also each fuzzy number, can fully and uniquely be represented by its α -cuts (Sec. 2.2); and

(2) α -cuts of each fuzzy number are closed intervals of real numbers for all $\alpha \in (0,1]$. These properties enable us to define arithmetic operations on fuzzy numbers in terms of arithmetic operations on their α -cuts (i.e., arithmetic operations on closed intervals). The latter operations are a subject of interval analysis, a well-established area of classical mathematics; we overview them in this section to facilitate our presentation of fuzzy arithmetic in the next section.

Let $*$ denote any of the four arithmetic operations on closed intervals: addition $+$, subtraction $-$, multiplication $\cdot$, and division $/$. Then,

$$[a, b] * [d, e] = \{f * g \mid a \leq f \leq b, d \leq g \leq e\}$$

is a general property of all arithmetic operations on closed intervals, except that $[a, b]/[d, e]$ is not defined when $0 \in [d, e]$. That is, the result of an arithmetic operation on closed intervals is again a closed interval.

The four arithmetic operations on closed intervals are defined as follows:

$$\begin{aligned} [a, b] + [d, e] &= [a + d, b + e], \\ [a, b] - [d, e] &= [a - e, b - d], \\ [a, b] \cdot [d, e] &= [\min(ad, ae, bd, be), \max(ad, ae, bd, be)], \end{aligned}$$

and, provided that $0 \notin [d, e]$,

$$\begin{aligned} [a, b]/[d, e] &= [a, b] \cdot [1/e, 1/d] \\ &= [\min(a/d, a/e, b/d, b/e), \max(a/d, a/e, b/d, b/e)]. \end{aligned}$$

Note that a real number r may also be regarded as a special (degenerated) interval $[r, r]$. When one of the intervals in (4.3) - (4.6) is degenerated, we obtain special operations; when both of them are degenerated, we obtain the standard arithmetic of real numbers.

The following are a few examples illustrating the interval-valued arithmetic operations defined by (4.3) – (4.6) :

$$\begin{aligned} [2,5] + [1,3] &= [3,8] & [0,1] + [-6,5] &= [-6,6], \\ [2,5] - [1,3] &= [-1,4] & [0,1] - [-6,5] &= [-5,7], \\ [-1,1] \cdot [-2, -0.5] &= [-2,2] & [3,4] \cdot [2,2] &= [6,8], \\ [-1,1]/[-2, -0.5] &= [-2,2] & [4,10]/[1,2] &= [2,10]. \end{aligned}$$

Arithmetic operations on closed intervals satisfy some useful properties. To overview them, let $A = [a_1, a_2]$, $B = [b_1, b_2]$, $C = [c_1, c_2]$, $0 = [0, 0]$, $1 = [1, 1]$. Using these symbols, the properties are formulated as follows:

1. $A + B = B + A$,
 $A \cdot B = B \cdot A$ (commutativity).
2. $(A + B) + C = A + (B + C)$
 $(A \cdot B) \cdot C = A \cdot (B \cdot C)$ (associativity).
3. $A = 0 + A = A + 0$
 $A = 1 \cdot A = A \cdot 1$ (identity)
4. $A \cdot (B + C) \subseteq A \cdot B + A \cdot C$ (subdistributivity).
5. If $b \cdot c \geq 0$ for every $b \in B$ and $c \in C$, then $A \cdot (B + C) = A \cdot B + A \cdot C$ (distributivity). Furthermore, if $A = [a, a]$, then $a \cdot (B + C) = a \cdot B + a \cdot C$.
6. $0 \in A - A$ and $1 \in A/A$.
7. If $A \subseteq E$ and $B \subseteq F$, then:
$$A + B \subseteq E + F$$

$$A - B \subseteq E - F$$

$$A \cdot B \subseteq E \cdot F$$

$$A/B \subseteq E/F$$
 (inclusion monotonicity).

2.4 Arithmetic operations on fuzzy numbers

There are two methods to perform fuzzy arithmetic.

(i) Based on interval arithmetic

Let A and B denote fuzzy numbers and let ‘*’ denote any of the four basic arithmetic

operations $\{+, -, \cdot, /\}$. Then we define a fuzzy set on $\mathbb{R}$, $A*B$ by defining its α -cut, ${}^\alpha(A*B)$

Where ${}^\alpha(A*B) = {}^\alpha A * {}^\alpha B$ for any $\alpha \in [0, 1]$

where $(A*B)^\alpha$ can be expressed as,

$$A*B = \bigcup_{\alpha \in (0,1]} \alpha(A * B)$$

Note:

* When $*$ = /, clearly, we have to require that $0 \notin B$ for all α in $(0,1]$.

* Since ${}^\alpha(A*B)$ is a closed interval for each $\alpha \in (0,1]$ and A, B are fuzzy numbers, $A*B$ is also a fuzzy number.

E.g.: consider two triangular- shape fuzzy numbers A and B defined as follows:

$$A(x) = \begin{cases} 0 & \text{for } x \leq -1 \text{ and } x > 3 \\ (x+1)/2 & \text{for } -1 < x < 1 \\ (3-x)/2 & \text{for } 1 < x \leq 3 \end{cases}$$

$$B(x) = \begin{cases} 0 & \text{for } x \leq 1 \text{ and } x > 5 \\ (x-1)/2 & \text{for } 1 < x \leq 3 \\ (5-x)/2 & \text{for } 3 < x \leq 5 \end{cases}$$

$${}^\alpha(A*B) = {}^\alpha A * {}^\alpha B$$

(i) ${}^\alpha(A+B)$

In order to find ${}^\alpha(A+B)$, it is enough to find ${}^\alpha A + {}^\alpha B$

To find ${}^\alpha A$:

$${}^\alpha A = \{x \mid A(x) \geq \alpha\}; \alpha \in (0,1]$$

$$\begin{aligned} {}^\alpha A &= \{(-1, 1] \mid (x+1)/2 \geq \alpha\} \\ &= \{(-1, 1] \mid x \geq 2\alpha - 1\} \dots (1) \end{aligned}$$

$$\begin{aligned} {}^\alpha A &= \{(-1, 1] \mid (3-x)/2 \geq \alpha\} \\ &= \{(1, 3] \mid -x \geq 2\alpha - 3\} \\ &= \{(1, 3] \mid x \leq 3 - 2\alpha\} \dots (2) \end{aligned}$$

From (1) and (2):

$${}^\alpha A = \{(-1, 3] \mid 2\alpha - 1 \leq x \leq 3 - 2\alpha\}$$

$${}^\alpha A = [2\alpha - 1, 3 - 2\alpha]$$

In the similar way:

$${}^\alpha B = [2\alpha + 1, 5 - 2\alpha]$$

$${}^\alpha (A + B) = {}^\alpha A + {}^\alpha B$$

$$= [2\alpha - 1, 3 - 2\alpha] + [2\alpha + 1, 5 - 2\alpha]$$

$$= [4\alpha, 8 - 4\alpha]$$

In the same way, we can find ${}^{\alpha}(A - B)$, ${}^{\alpha}(A \cdot B)$, ${}^{\alpha}(A/B)$

Extension principle:

Any given function $f: X \rightarrow Y$ induces two functions, $f: F(X) \rightarrow F(Y)$,

$$f^{-1}: F(Y) \rightarrow F(X)$$

which are defined by,

$$[f(A)](y) = \sup A(x) \quad \forall A \in F(X) \text{ and}$$

$$[f^{-1}(B)](x) = B(f(x)) \quad \forall B \in F(Y)$$

Fuzzy arithmetic based on the extension principle

Let ‘*’ denote any of the four basic arithmetic operations $\{+, -, \cdot, /\}$.

Let A, B denote fuzzy numbers.

Then $A*B$, a fuzzy set defined by the equation, on R is

$$(A * B)(z) = \sup_{z = x * y} \min[A(x), B(y)] \quad \forall z \in R$$

Specifically, for all $z \in R$, we define:

$$(A + B)(z) = \sup_{z = x + y} \min[A(x), B(y)]$$

$$(A - B)(z) = \sup_{z = x - y} \min[A(x), B(y)]$$

$$(A \cdot B)(z) = \sup_{z = x \cdot y} \min[A(x), B(y)]$$

$$(A / B)(z) = \sup_{z = x / y} \min[A(x), B(y)]$$

Theorem: Let $* \in \{+, -, \cdot, /\}$, and let A, B denote continuous fuzzy numbers.

Then, the fuzzy set $A * B$ defined by (4.9) is a continuous fuzzy number.

Proof:

First, we prove (4.7) by showing that ${}^{\alpha}(A * B)$ is a closed interval for every

$\alpha \in (0, 1]$. For any $z \in {}^{\alpha}A * {}^{\alpha}B$, there exist some $x_0 \in {}^{\alpha}A$ and $y_0 \in {}^{\alpha}B$ such that $z = x_0 * y_0$.

Thus,

$$\begin{aligned}(A * B)(z) &= \sup_{z=x*y} \min[A(x), B(y)] \\ &\geq \min[A(x_0), B(y_0)] \\ &\geq \alpha\end{aligned}$$

Hence, $z \in {}^\alpha(A * B)$ and, consequently,

$${}^\alpha A * {}^\alpha B \subseteq {}^\alpha(A * B).$$

For any $z \in {}^\alpha(A * B)$, we have

$$(A * B)(z) = \sup_{z=x*y} \min[A(x), B(y)] \geq \alpha.$$

Moreover, for any $n > [1/\alpha] + 1$, where $[1/\alpha]$ denotes the largest integer that is less than or equal to $1/\alpha$, there exist x_n and y_n such that $z = x_n * y_n$ and

$$\min[A(x_n), B(y_n)] > \alpha - \frac{1}{n}.$$

That is, $x_n \in {}^{\alpha-1/n}A$, $y_n \in {}^{\alpha-1/n}B$ and we may consider two sequences, $\{x_n\}$ and $\{y_n\}$. Since

$$\alpha - \frac{1}{n} \leq \alpha - \frac{1}{n+1}$$

we have

$${}^{\alpha-1/(n+1)}A \subseteq {}^{\alpha-1/n}A, {}^{\alpha-1/(n+1)}B \subseteq {}^{\alpha-1/n}B$$

Hence, $\{x_n\}$ and $\{y_n\}$ fall into some ${}^{\alpha-1/n}A$ and ${}^{\alpha-1/n}B$, respectively. Since the latter are closed intervals, $\{x_n\}$ and $\{y_n\}$ are bounded sequences. Thus, there exists a convergent subsequence $\{x_{n,i}\}$ such that $x_{n,i} \rightarrow x_0$. To the corresponding subsequence $\{y_{n,i}\}$, there also exists a convergent subsequence $\{y_{n,i,j}\}$ such that $y_{n,i,j} \rightarrow y_0$. If we take the corresponding subsequence, $\{x_{n,i,j}\}$, from $\{x_{n,i}\}$, then $x_{n,i,j} \rightarrow x_0$. Thus, we have two sequences, $\{x_{n,i,j}\}$ and $\{y_{n,i,j}\}$, such that $x_{n,i,j} \rightarrow x_0$, $y_{n,i,j} \rightarrow y_0$, and $x_{n,i,j} * y_{n,i,j} = z$.

Now, since $*$ is continuous,

$$z = \lim_{j \rightarrow \infty} x_{n,i,j} * y_{n,i,j} = \left(\lim_{j \rightarrow \infty} x_{n,i,j} \right) * \left(\lim_{j \rightarrow \infty} y_{n,i,j} \right) = x_0 * y_0.$$

Also, since $A(x_{n,i,j}) > \alpha - \frac{1}{n_{i,j}}$ and $B(y_{n,i,j}) > \alpha - \frac{1}{n_{i,j}}$,

$$A(x_0) = A\left(\lim_{j \rightarrow \infty} x_{n,i,j}\right) = \lim_{j \rightarrow \infty} A(x_{n,i,j}) \geq \lim_{j \rightarrow \infty} \left(\alpha - \frac{1}{n_{i,j}}\right) = \alpha$$

and

$$B(y_0) = B\left(\lim_{j \rightarrow \infty} y_{n,i,j}\right) = \lim_{j \rightarrow \infty} B(y_{n,i,j}) \geq \lim_{j \rightarrow \infty} \left(\alpha - \frac{1}{n_{i,j}}\right) = \alpha.$$

Therefore, there exist $x_0 \in {}^\alpha A, y_0 \in {}^\alpha B$ such that $z = x_0 * y_0$. That is, $z \in {}^\alpha A * {}^\alpha B$. Thus,

$${}^\alpha(A * B) \subseteq {}^\alpha A * {}^\alpha B,$$

and, consequently,

$${}^\alpha(A * B) = {}^\alpha A * {}^\alpha B$$

Now we prove that $A * B$ must be continuous. Assume $A * B$ is not continuous at z_0 ; that is,

$$\lim_{z \rightarrow z_0^-} (A * B)(z) < (A * B)(z_0) = \sup_{z_0 = x * y} \min[A(x), B(y)].$$

Then, there must exist x_0 and y_0 such that $z_0 = x_0 * y_0$ and

$$\lim_{z \rightarrow z_0^-} (A * B)(z) < \min[A(x_0), B(y_0)]$$

Since the operation $* \in \{+, -, \cdot, /\}$ is monotonic with respect to the first and the second arguments, respectively, we can always find two sequences $\{x_n\}$ and $\{y_n\}$ such that $x_n \rightarrow x_0, y_n \rightarrow y_0$ as $n \rightarrow \infty$, and $x_n * y_n < z_0$ for any n . Let $z_n = x_n * y_n$; then $z_n \rightarrow z_0$ as $n \rightarrow \infty$. Thus,

$$\begin{aligned} \lim_{z \rightarrow z_0^-} (A * B)(z) &= \lim_{n \rightarrow \infty} (A * B)(z_n) = \lim_{n \rightarrow \infty} \sup_{z_n = x * y} \min[A(x), B(y)] \\ &\geq \lim_{n \rightarrow \infty} \min[A(x_n), B(y_n)] = \min\left[A\left(\lim_{n \rightarrow \infty} x_n\right), B\left(\lim_{n \rightarrow \infty} y_n\right)\right] = \min[A(x_0), B(y_0)]. \end{aligned}$$

This contradicts (4.14) and, therefore, $A * B$ must be a continuous fuzzy number. This completes the proof.

MODULE 3: FUZZY RELATIONS AND FUZZY GRAPHS

3.1 Fuzzy Relations on Sets and Fuzzy Sets

Fuzzy relations are fuzzy subsets of $X \times Y$, that is, mappings from $X \rightarrow Y$. They have been studied by a number of authors, in particular by Zadeh [1965, 1971], Kaufmann [1975], and Rosenfeld [1975]. Applications of fuzzy relations are widespread and important. We shall consider some of them and point to more possible uses at the end of this chapter. We shall exemplarily consider only binary relations. A generalization to n -ary relations is straightforward.

Definition

Let $X, Y \subseteq \mathbb{R}$ be universal sets; then

$$\tilde{R} = \{((x, y), \mu_{\tilde{R}}(x, y)) \mid (x, y) \in X \times Y\}$$

is called a fuzzy relation on $X \times Y$.

Example

Let $X = Y = \mathbb{R}$ and $\tilde{R} :=$ "considerably larger than." The membership function of the fuzzy relation, which is, of course, a fuzzy set on $X \times Y$, can then be

$$\mu_{\tilde{R}}(x, y) = \begin{cases} 0 & \text{for } x \leq y \\ \frac{(x - y)}{10y} & \text{for } y < x \leq 11y \\ 1 & \text{for } x > 11y \end{cases}$$

A different membership function for this relation could be

$$\mu_{\tilde{R}}(x, y) = \begin{cases} 0 & \text{for } x \leq y \\ (1 + (y - x)^{-2})^{-1} & \text{for } x > y \end{cases}$$

For discrete supports, fuzzy relations can also be defined by matrixes.

Example

	y_1	y_2	y_3	y_4
$\tilde{R} = "x \text{ considerably larger than } y": x_2$	.8	1	.1	.7
	0	.8	0	0
x_3	.9	1	.7	.8

Let $X = \{x_1, x_2, x_3\}$ and $Y = \{y_1, y_2, y_3, y_4\}$

	y_1	y_2	y_3	y_4
$\tilde{Z} = "y \text{ very close to } x": x_2$	.4	0	.9	.6
	.9	.4	.5	.7
	.3	0	.8	.5

and

In definition it was assumed that $\mu_{\tilde{R}}$ was a mapping from $X \times Y$ to $[0,1]$; that is, the definition assigns to each pair (x, y) a degree of membership in the unit interval. In some instances, such as in graph theory, it is useful to consider fuzzy relations that map from fuzzy sets contained in the universal sets into the unit interval. Then definition 6-1 has to be generalized [Rosenfeld 1975].

Definition

Let $X, Y \subseteq \mathbb{R}$ and

$$\begin{aligned}\tilde{A} &= \{(x, \mu_{\tilde{A}}(x)) \mid x \in X\}, \\ \tilde{B} &= \{(y, \mu_{\tilde{B}}(y)) \mid y \in Y\}, \text{ two fuzzy sets.}\end{aligned}$$

Then $\tilde{R} = \{[(x, y), \mu_{\tilde{R}}(x, y)] \mid (x, y) \in X \times Y\}$ is a fuzzy relation on $\tilde{A}$ and $\tilde{B}$ if

$$\mu_{\tilde{R}}(x, y) \leq \mu_{\tilde{A}}(x), \forall (x, y) \in X \times Y$$

and

$$\mu_{\tilde{R}}(x, y) \leq \mu_{\tilde{B}}(y), \forall (x, y) \in X \times Y.$$

This definition will be particularly useful when defining fuzzy graphs: Let the elements of the fuzzy relation of definition 6-2 be the nodes of a fuzzy graph that is represented by this fuzzy relation. The degrees of membership of the elements of the related fuzzy sets define the "strength" of or the flow in the respective nodes of the graph, while the degrees of membership of the corresponding pairs in the relation are the "flows" or "capacities" of the edges. The additional requirement of definition 6-2 ($\mu_{\tilde{R}}(x, y) \leq \min\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(y)\}$) then ensures that the "flows" in the edges of the graph can never exceed the flows in the respective nodes.

Fuzzy relations are obviously fuzzy sets in product spaces. Therefore set theoretic and algebraic operations can be defined for them in analogy to the definitions in chapters 2 and 3 by utilizing the extension principle.

Definition

Let $\tilde{R}$ and $\tilde{Z}$ be two fuzzy relations in the same product space. The union/intersection of $\tilde{R}$ with $\tilde{Z}$ is then defined by

$$\begin{aligned}\mu_{\tilde{R} \cup \tilde{Z}}(x, y) &= \max\{\mu_{\tilde{R}}(x, y), \mu_{\tilde{Z}}(x, y)\}, (x, y) \in X \times Y \\ \mu_{\tilde{R} \cap \tilde{Z}}(x, y) &= \min\{\mu_{\tilde{R}}(x, y), \mu_{\tilde{Z}}(x, y)\}, (x, y) \in X \times Y\end{aligned}$$

3.2 Compositions of Fuzzy Relations

Fuzzy relations in different product spaces can be combined with each other by the operation "composition." Different versions of "composition" have been suggested, which differ in their results and also with respect to their mathematical properties. The max-min composition has become the best known and the most frequently used one. However, often the so-called max-product or max-average compositions lead to results that are more appealing.

Definition

Max-min composition: Let $\tilde{R}_1(x, y), (x, y) \in X \times Y$ and $\tilde{R}_2(y, z), (y, z) \in Y \times Z$ be two fuzzy relations. The max-min composition $\tilde{R}_1 \max - \min \tilde{R}_2$ is then the fuzzy set

$$\tilde{R}_1 \circ \tilde{R}_2 = \left\{ \left[(x, z), \max_y \left\{ \min\{\mu_{\tilde{R}_1}(x, y), \mu_{\tilde{R}_2}(y, z)\} \right\} \right] \mid x \in X, y \in Y, z \in Z \right\}$$

$\mu_{\tilde{R}_1 \circ \tilde{R}_2}$ is again the membership function of a fuzzy relation on fuzzy sets (definition 6-2).

A more general definition of composition is the "max-* composition."

Definition

Let $\tilde{R}_1$ and $\tilde{R}_2$ be defined as in definition. The max-* composition of $\tilde{R}_1$ and $\tilde{R}_2$ is then defined as

$$\tilde{R}_1 * \tilde{R}_2 = \{[(x, z), \max(\mu_{\tilde{R}_1}(x, y) * \mu_{\tilde{R}_2}(y, z))] \mid x \in X, y \in Y, z \in Z\}$$

If * is an associative operation that is monotonically nondecreasing in each argument, then the max-* composition corresponds essentially to the max-min composition. Two special cases of the max-* composition are proposed in the next definition.

Properties of the Min-Max Composition

Associativity

- The min-max composition is associative: $(R_3 \circ R_2) \circ R_1 = R_3 \circ (R_2 \circ R_1)$
- Hence, $R_1 \circ R_1 \circ R_1 = R_1^{\wedge 3}$, and the third power of a fuzzy relation is defined.

Reflexivity

- A fuzzy relation R in $X \times X$ is:
 - Reflexive if $R(x, x) = 1$ for all $x \in X$
 - ε -reflexive if $R(x, x) \geq \varepsilon$ for all $x \in X$
 - Weakly reflexive if $R(x, y) \leq \min(R(x, x), R(y, y))$ for all $x, y \in X$

Symmetry

- A fuzzy relation R is:
 - Symmetric if $R(x, y) = R(y, x)$ for all $x, y \in X$
 - Antisymmetric if for $x \neq y$, either $R(x, y) \neq R(y, x)$ or $R(x, y) = R(y, x) = 0$
 - Perfectly antisymmetric if for $x \neq y$, whenever $R(x, y) > 0$, then $R(y, x) = 0$

Transitivity

- A fuzzy relation R is (max-min) transitive if $R \circ R \subseteq R$

Remark 1 (Properties for Max-Min Compositions)

1. If R_1 is reflexive and R_2 is arbitrary, then $R_1 \circ R_2 \supseteq R_2$ and $R_2 \circ R_1 \supseteq R_2$
2. If R is reflexive, then $R \subseteq R \circ R$
3. If R_1 and R_2 are reflexive, so is $R_1 \circ R_2$
4. If R_1 and R_2 are symmetric, then $R_1 \circ R_2$ is symmetric if $R_1 \circ R_2 = R_2 \circ R_1$
5. If R is symmetric, so is each power of R

Remark 2 (Combinations of Properties)

1. If R is symmetric and transitive, then $R(x, y) \geq R(x, x)$ for all $x, y \in X$
2. If R is reflexive and transitive, then $R \circ R = R$
3. If R_1 and R_2 are transitive and $R_1 \circ R_2 = R_2 \circ R_1$, then $R_1 \circ R_2$ is transitive

Note: These properties hold for the max-min composition. Some properties may hold for other compositions as well.

3.3 Fuzzy Graphs

It was already mentioned that definitions and of a fuzzy relation can also be interpreted as defining a fuzzy graph. In order to stay in line with the terminology of traditional graph theory we shall use the following definition of a fuzzy graph.

Definition

Let E be the (crisp) set of nodes. A fuzzy graph is then defined by

$$\tilde{G}(x_i, x_j) = \left\{ \left((x_i, x_j), \mu_{\tilde{G}}(x_i, x_j) \right) \mid (x_i, x_j) \in E \times E \right\}$$

If $\tilde{E}$ is a fuzzy set, a fuzzy graph would have to be defined in analogy to definition 6-2.

Example

a. Let $E = \{A, B, C\}$.

Considering only three possible degrees of membership, graphs could be described as shown in figure 6-1.

b. Let $E = \{x_1, x_2, x_3, x_4\}$; then a fuzzy graph could be described as

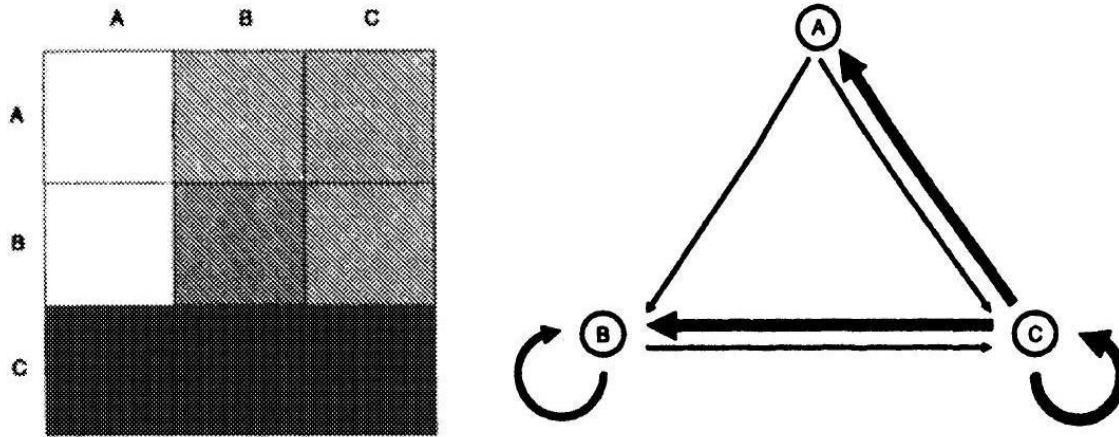


Figure 6-1. Fuzzy graphs.

$$\begin{aligned} \tilde{G}(x_i, x_j) = \{ & [(x_1, x_2), .3], [(x_1, x_3), .6], [(x_1, x_1), 1], \\ & [(x_2, x_1), .4], [(x_3, x_1), .2], [(x_3, x_2), .5], \\ & [(x_4, x_3), .8] \} \end{aligned}$$

Example shows directed fuzzy binary graphs. Graphs can, of course, also be defined in higher-dimension product spaces. We shall, however, focus our attention on finite undirected binary graphs; that is, we shall assume in the following that the fuzzy relation representing a graph is symmetric. The arcs can then be considered as unordered pairs of nodes. In analogy to traditional graph theory, fuzzy graph theoretic concepts can be defined.

Definition

$\tilde{H}(x_i, x_j)$ is a fuzzy subgraph of $\tilde{G}(x_i, x_j)$ if

$$\mu_{\tilde{H}}(x_i, x_j) \leq \mu_{\tilde{G}}(x_i, x_j) \forall (x_i, x_j) \in E \times E$$

$\tilde{H}(x_i, x_j)$ spans graph $\tilde{G}(x_i, x_j)$ if the node sets of $\tilde{H}(x_i, x_j)$ and $\tilde{G}(x_i, x_j)$ are equal, that is, if they differ only in their arc weights.

Example

Let $\tilde{G}(x_i, x_j)$ be defined as in example 6-11b. A spanning subgraph of $\tilde{G}(x_i, x_j)$ is then

$$\tilde{H}(x_i, x_j) = \{[(x_1, x_2), .2], [(x_1, x_3), .4], [(x_3, x_2), .4] \\ [(x_4, x_3), .7]\}$$

Definition

A path in a fuzzy graph $\tilde{G}(x_i, x_j)$ is a sequence of distinct nodes, $x_0, x_1, \dots, x_n$, such that for all $(x_i, x_{i+1}), \mu_{\tilde{G}}(x_i, x_{i+1}) > 0$. The strength of the path is $\min\{\mu_{\tilde{G}}(x_i, x_{i+1})\}$ for all nodes contained in the path. The length of a path $n > 0$ is the number of nodes contained in the path. Each pair of nodes $(x_i, x_{i+1}), \mu(x_i, x_{i+1}) > 0$ is called an edge (arc) of the graph. A path is called a cycle if $x_0 = x_n$ and $n \geq 3$.

It would be straightforward to call the length of the shortest path between two nodes of the graph the distance between these nodes. This definition, however, has some disadvantages. It is therefore more reasonable to define the distance between two nodes as follows [Rosenfeld 1975, p. 58]:

Definition

The μ -length of a path $p = x_0, \dots, x_n$ is equal to

$$L(p) = \sum_{i=1}^n \frac{1}{\mu(x_i, x_{i+1})}$$

The μ -distance $d(x_i, x_j)$ between two nodes x_i, x_j is the smallest μ -length of any path from x_i to $x_j, x_i, x_j \in \tilde{G}$.

It can then be shown [see Rosenfeld 1975, p. 88] that $d(x_i, x_j)$ is a metric (in undirected graphs!).

Definition

Two nodes that are joined by a path are called connected nodes.

Connectedness is a relation that is also transitive.

Definition

A fuzzy graph is a forest if it has no cycles; that is, it is an acyclic fuzzy graph. If the fuzzy forest is connected, it is called a tree. (A fuzzy graph that is a forest has to be distinguished from a fuzzy graph that is a fuzzy forest.) The latter shall not be discussed here [see Rosenfeld 1975, p. 92].

Example

The fuzzy graphs shown in figure 6-2 are forests. The graphs shown in figure 6-3 are not.

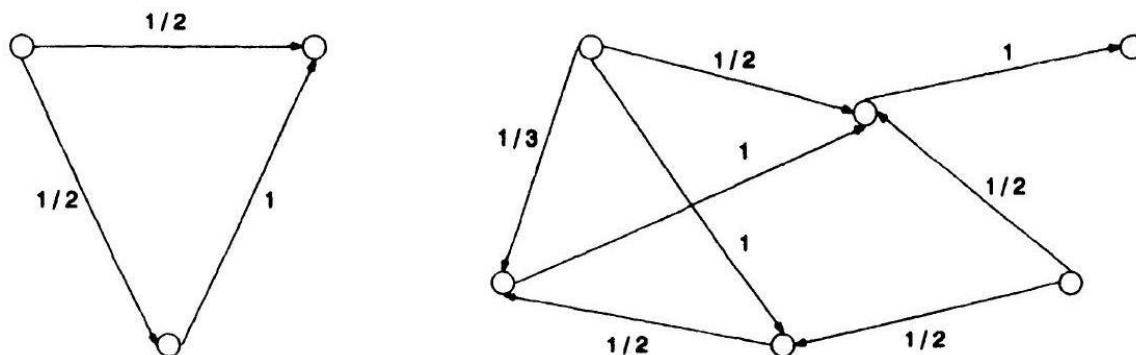


Figure 6-2. Fuzzy forests.

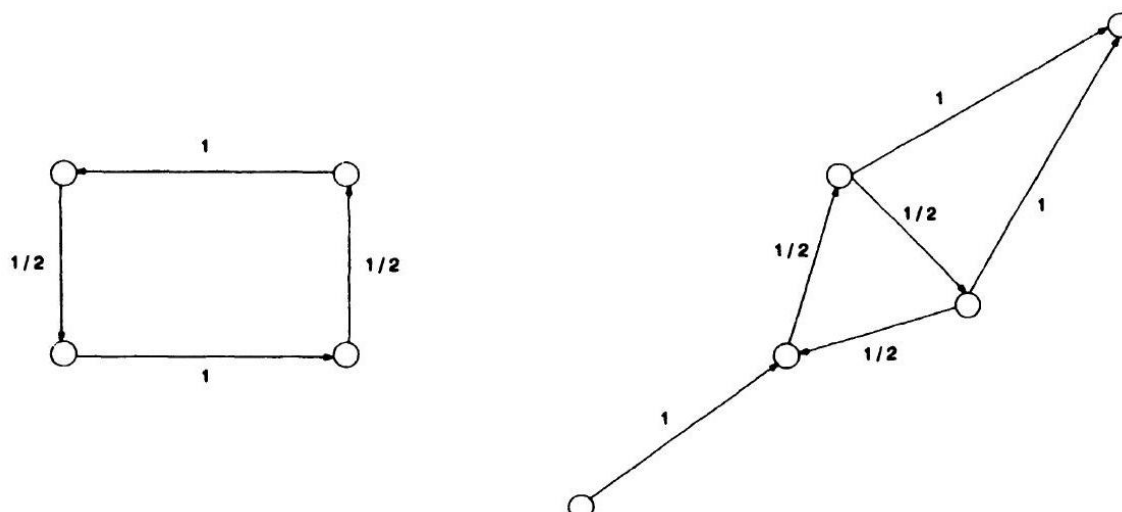


Figure 6-3. Graphs that are not forests.

3.4 Special Fuzzy Relations

Relations that are of particular interest to us are fuzzy relations that pertain to the similarity of fuzzy sets and those that order fuzzy sets. All of the relations discussed below are reflexive, that is, $\mu_{\tilde{R}}(x, x) = 1 \forall x \in X$ [Zadeh 1971], and they are max-min transitive, that is, $\tilde{R} \circ \tilde{R} \subseteq \tilde{R}$ or $\mu_{\tilde{R}}(x, z) \geq \min\{\mu_{\tilde{R}}(x, y), \mu_{\tilde{R}}(y, z)\} \forall x, y, z \in X$. It should be noted that other kinds of transitivity have been defined [see Bezdek and Harris 1978]. These, however, will not be discussed here. The main difference between similarity relations and order relations is the property of symmetry or antisymmetry, respectively.

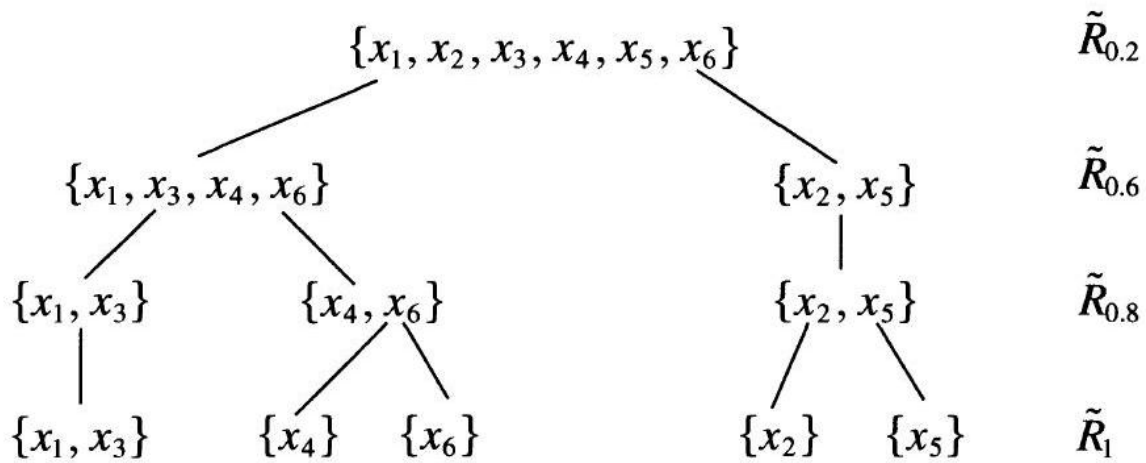
Definition: A similarity relation is a fuzzy relation $\mu_s(\cdot)$ that is reflexive, symmetrical, and max-min transitive.

Example

The following relation is a similarity relation [Zadeh 1971]:

	x_1	x_2	x_3	x_4	x_5	x_6
x_1	1	.2	1	.6	.2	.6
x_2	.2	1	.2	.2	.8	.2
$\tilde{R}_s: x_3$	1	.2	1	.6	.2	.6
x_4	.6	.2	.6	1	.2	.8
x_5	.2	.8	.2	.2	1	.2
x_6	.6	.2	.6	.8	.2	1

A similarity relation of a finite number of elements can also be represented by a similarity tree, similar to a dendogram. In this tree, each level represents an α cut (α -level set) of the similarity relation. For the above similarity relation, the similarity tree is shown below. The sets of elements on specific α -levels can be considered as similarity classes of α -level.



The properties of a similarity relation as defined in definition 6-20 are rather restrictive and not quite in accordance with fuzzy set thinking: Reflexitivity could be considered as being too restrictive and hence weakened by substituting

these requirements by ε -reflexivity or weak reflexivity (cf. definition 6-10). The max-min transitivity can be replaced by any max-* transitivity listed in definition 6-10 or in remark 6-1.

We shall now turn to fuzzy order relations: As already mentioned, similarity relations and order relations are primarily distinguished by their degree of symmetry.

Roughly speaking, similarity relations are fuzzy relations that are reflexive, (max-min) transitive, and symmetrical; order relations, however, are not symmetrical. To be more precise, even different kinds of fuzzy order relations differ by their degree of symmetry.

A fuzzy relation that is (max-min) transitive and reflexive is called a *fuzzy preorder relation*.

Definition

A fuzzy relation that is (min-max) transitive, reflexive, and antisymmetric is called a *fuzzy order relation*. If the relation is perfectly antisymmetric, it is called a *perfect fuzzy order relation* [Kaufmann 1975, p. 113]. It is also called a *fuzzy partial order relation* [Zadeh 1971].

Fuzzy Preorder Relations

A fuzzy relation that is max-min transitive and reflexive is called a fuzzy preorder relation. Fuzzy preorder relations are less restrictive compared to similarity relations as they don't require symmetry.

Fuzzy Order Relations

A fuzzy relation that is min-max transitive, reflexive, and antisymmetric is called a fuzzy order relation. If the relation is perfectly antisymmetric, it is called a perfect fuzzy order relation or fuzzy partial order relation.

A total fuzzy order relation or fuzzy linear ordering is a fuzzy order relation such that for any elements x and y , either $R(x, y) > 0$ or $R(y, x) > 0$.

Fuzzy order relations introduce the concept of antisymmetry, making them suitable for representing partial orders and rankings.

These special fuzzy relations play a crucial role in various applications, providing a framework to model and analyse relationships in a more flexible and nuanced manner than traditional crisp relations.

MODULE 4: FUZZY LOGIC

4.1 Classical Logic: An overview

In this chapter, we provide a brief overview of the basic concepts of fuzzy logic. It is solely intended to introduce terminology and notation employed in our discussion of fuzzy logic. To provide a foundation for understanding fuzzy logic, we begin with a brief review of classical logic.

Logic is the study of the methods and principles of reasoning in all of its possible forms. Classical logic deals with propositions that are required to be either true or false.

Basic definitions

Proposition:

Propositions are declarative sentences, either true or False but not both. Symbolically, propositions are denoted by the letters $p, q, r, \dots$ etc. are called propositional variables.

Negation of a proposition:

Denial of a statement or proposition is called negation. It is denoted by ' $\neg p$ ' or ' $\sim p$ ' or ' $\bar{p}$ '

Conjunction:

Two simple propositions connected by the word 'and' is said to form a conjunction. A conjunction formed by p and q is denoted by ' $p \wedge q$ '.

Disjunction:

Two simple propositions connected by the word 'or' is said to form a disjunction. It is denoted by ' $p \vee q$ '.

Exclusive or:

If p and q are two proportions then proposition is true when exactly one of p and, q is true and is false otherwise. It is dented by ' $p \oplus q$ '

Conditional statement:

Two simple propositions connected by the phrase 'If -then' is said to form a conditional statement.

Bi- Conditional statement:

Two simple statements connected by the phrase 'iff' is called a bi-conditional statement.

Tautology and contradiction:

A tautology is a proposition, which is always true (T).

A contradiction is a proposition, which is always false (F).

Algebra of propositions	
$p \wedge p = p$	Idem potent law
$p \wedge q = q \wedge p$	Commutative Law
$(p \wedge q) \wedge r = p \wedge (q \wedge r)$	Associative Law
$p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$	Distributive Law
$(a) p \wedge T = p$ $(b) p \vee F = p$ $(c) p \vee T = T$ $(d) p \wedge F = F$	Identity Laws
$(a) \sim T = F$ $(b) \sim F = T$ $(c) \sim(\sim p) = p$ $(d) p \wedge \sim p = F$ $p \vee \sim p = T$ (e)	Compliment Laws
$\sim(p \vee q) = (\sim p) \wedge (\sim q)$	Demorgan's Law
$\sim(p \leftrightarrow q) = p \leftrightarrow \sim q$ or $\sim p \leftrightarrow q$	Demorgan's Law
$p \wedge (p \vee q) = p$	Absorption Laws
$\sim(\sim p) = p$	Double negation Law

Propositional function

Let A be a set, then an expression of the form $p(x)$ is said to be a propositional function or an open sentence on A if $p(a)$ is a statement for all $a \in A$.

Eg. $P(x) = x + 2 (> 5)$ is a propositional function on the set $A = \{1, 2, 3, 4, 5\}$.

Subject and predicate

In a propositional function, the variable part is called the subject and the conditional part is called the predicate.

Quantifiers

Propositional functions become statements when they are assigned values. However, there is another important concept called quantification, to create a proposition from a propositional function. The words "all," "some," "many," and "none" are used in quantification. Such words are called quantifiers.

There are two kinds of quantification:

1. **Universal quantifier:** The symbol $\forall$, which reads "for all" or "all," is called a universal quantifier.
2. **Existential quantifier:** The symbol $\exists$, which reads "there exists" or "some," is called an existential quantifier.

Argument

An argument is a sequence of statements which end with a conclusion. Therefore, it has truth-values. If it is true, it is called a valid argument. If it is false, it is called a fallacy.

Rules of Inference

We can always use truth table to show that an argument is valid, But it is a tedious approach when an argument involves large number of propositional variables (e.g. if there are 2^{10} different rows for the truth table). To overcome this difficulty, we establish the validity of some relatively simple argument forms, called rules of inference.

No.	Name	Rule	Conclusion
1	Modus Ponens	$[(p \rightarrow q) \wedge p] \rightarrow q$	q
2	Modus Tollens	$[\neg q \wedge (p \rightarrow q)] \rightarrow \neg p$	$\neg p$
3	Syllogism	$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow p \rightarrow r$	$p \rightarrow r$
4	Disjunctive Syllogism	$[(p \vee q) \wedge \neg p] \rightarrow q$	q
5	Addition	$p \rightarrow (p \vee q)$	$p \vee q$
6	Simplification	$p \wedge q \rightarrow p$	p
7	Conjunction	$p \text{ and } q \rightarrow p \wedge q$	$p \wedge q$
8	Resolution	$[(p \vee q) \wedge (\neg p \vee r)] \rightarrow q \vee r$	$q \vee r$

4.2 Fuzzy logic

The basic assumption upon which classical logic is based on the idea that every proposition is either true or false. But propositions about future events are neither actually true nor actually false, but potentially either; hence their truth value is undetermined least prior to the event. The classical two-valued logic can be extended into three-valued logic in various ways, it is common in these logics to denote the truth, falsity, and indeterminacy by 1, 0, and 1/2, respectively.

The logical connectives of negation, disjunction, conjunction, and implication for fuzzy logic are also defined.

If P and Q are two simple propositions defined on fuzzy sets A and B, then

(i) Negation

$$\bar{P} = 1 - P$$

(ii) Disjunction

$$p \vee q : x \text{ is } A \text{ or } B$$

$$p \vee q = \max(P, Q)$$

(iii) Conjunction

$$p \wedge q : x \text{ is } A \text{ and } B$$

$$p \wedge q = \min(P, Q)$$

(iv) Implication

$p \rightarrow q$: x is A then x is B

$$p \rightarrow q = \min(1, 1 + b - a) \text{ [Lukasiewicz implication]}$$

(v) Two Implication

$$P \leftrightarrow q = 1 - |a - b|$$

4.3 Fuzzy Propositions

The fundamental difference between classical propositions and Fuzzy propositions is in the range of their truth values. While each classical proposition is required to be either true or false, the truth or falsity of Fuzzy propositions is a matter of degree. Assuming that truth and falsity are expressed by numbers 1 and 0 respectively, the degree of truth of each fuzzy proposition is expressed by a number in the unit interval [0, 1].

Fuzzy propositions are classified into four types:

1. Unconditional and unqualified propositions
2. Unconditional and qualified propositions
3. Conditional and unqualified propositions
4. Conditional and qualified propositions

1. Unconditional and Unqualified Fuzzy Propositions.

The fuzzy proposition of this type, denoted as P, is expressed by the sentence:

$$P: A \text{ is } F,$$

Where 'A' is a variable that takes value 'a' from some universal set 'x', and F is a fuzzy predicate such as tall, expensive, high, low, or normal and so on. Give a

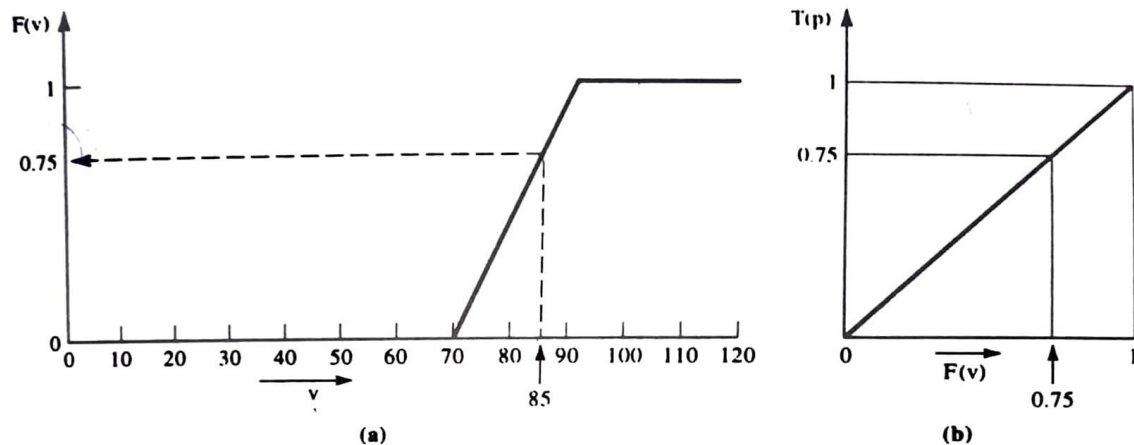
Particular value of 'A' (say a, this value belongs to F with membership grade F(a). Then the degree of truth, T(P) of proposition 'P' is

$$T(P) = F(a)$$

For example, let 'A' represent the air temperature at some particular place on Earth (measured in Fahrenheit), and let's define the membership function for the predicate "high" as shown in the figure.

Then the fuzzy proposition 'P' is expressed as:

P: "Temperature(A) is high(F)".



Components of the fuzzy proposition *p*: Temperature (*v*) is high (*F*).

Here $a=85$, the $F(85)=0.75$ and $T(P)=0.75$

2.Unconditional and qualified propositions

Propositions of this type are characterized by either the truth value qualifier **F** or the probability qualifier **P** of the form

P: A is F is B

- **A** and **F** have the same meaning as above
- **B** is a fuzzy truth qualifier, and
- **P** is a fuzzy probability qualifier.

In general, the degree of truth, **T(P)**, of any truth-qualified proposition **P** is:

$$T(P) = F(a)$$

Example:

Consider the proposition **P**:

P: Tina (A) is young (F) is very true (B)

Assume that Tina's age is 26. Then, from the following graph:

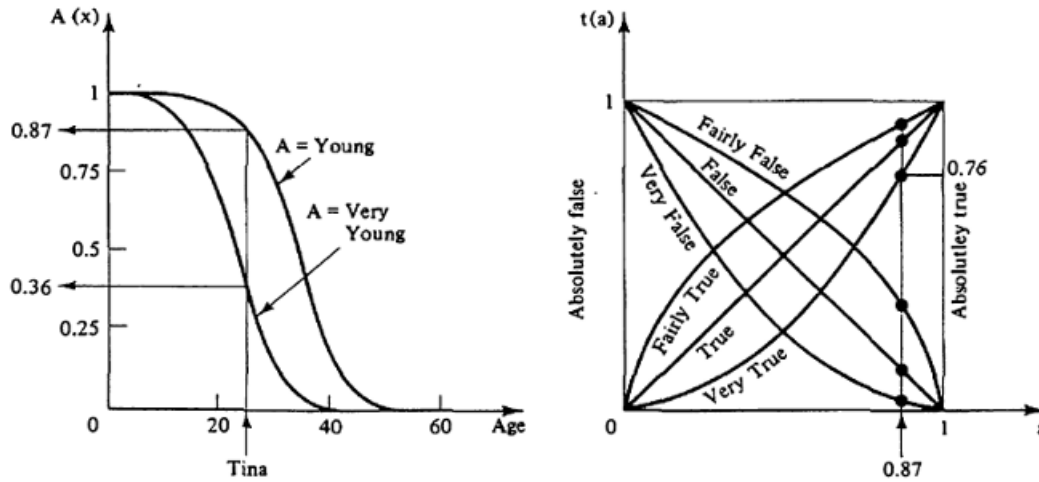


Figure . . . Truth values of a fuzzy proposition

she belongs to the set representing "young" with membership grade function **0.87**.

Hence, our proposition belongs to the set of propositions that are very true with membership grade function **0.76**.

i.e, $a = 26$, then $F(26) = 0.87$ and $T(P) = B(F(26)) = 0.76$

3. Conditional and Unqualified Fuzzy Propositions

Propositions of this type are expressed by the form

P: If X is A, then Y is B,

where X and Y are variables, whose values are in sets X and Y, respectively, and A and B are fuzzy sets on X and Y, respectively. These propositions may also be viewed as propositions of the form

$\langle X, Y \rangle$ is R,

where R is a fuzzy set on $X \times Y$ that is determined for each $x \in X$ and each $y \in Y$ by the formula

$$R(x,y) = J[A(x),B(y)],$$

where J denotes a binary operation on [0,1] representing a suitable fuzzy implication.

$$J(a,b) = \min(1, 1-a+b)$$

Example

Let $A = \{(x_1, 0.1), (x_2, 0.8), (x_3, 1)\},$

$$B = \{(y_1, 0.5), (y_2, 1)\}$$

Then our proposition is, "If x is A then y is B"

$$R(x, y) = J(a, b) = \min(1, 1 - a + b)$$

$$R(x_1, y_1) = \min(1, 1 - 0.1 + 0.5)$$

$$= \min(1, 1.4)$$

$$= 1$$

$$R(x_1, y_2) = \min(1, 1 - a + b)$$

$$= \min(1, 1 - 0.1 + 1)$$

$$= \min(1, 1.9)$$

$$= 1$$

Similarly

$$R(x_2, y_1) = 0.7$$

$$R(x_2, y_2) = 1$$

$$R(x_3, y_1) = 0.5$$

$$R(x_3, y_2) = 1$$

$$\Rightarrow R = \{(x_1, y_1, 1), (x_1, y_2, 1), (x_2, y_2, 1), (x_3, y_2), (x_2, y_1, 0.7), (x_3, y_1, 0.5)\}$$

$$\Rightarrow T(P) = 1 \text{ when } x = x_1 \quad y = y_1$$

$$x = x_1 \quad y = y_2$$

$$x = x_2 \quad y = y_2$$

$$x = x_3 \quad y = y_2$$

$$T(P) = 0.7 \text{ when } x = x_2, y = y_1$$

$$T(P) = 0.5 \text{ when } x = x_3, y = y_1$$

4. Conditional and Qualified propositions

Propositions of this Type can be expressed as,

$$P: x \text{ is } A \text{ then } y \text{ is } B \text{ is } S$$

Since methods introduced for the other types of proposition can be combined to deal with propositions of this type, we do not deem it necessary to discuss them later.

4.4 Fuzzy Quantifiers

Fuzzy quantifiers are fuzzy numbers that take part in fuzzy propositions. They are of two kinds,

1. Absolute quantifiers
2. Relative quantifiers.

1. Absolute quantifiers:

Absolute quantifiers are defined on R and characterize linguistic terms such as about 10, much more than 100, at least about 5, and so on.

There are two basic Forms of propositions that contain absolute quantifiers.

The First form is,

$$P: \text{There are } Qi's \text{ in } I \text{ such that } V(i) \text{ is } F$$

where V is a variable that for each Individual 'i' in a given set I, assumes a value V(i), F is a fuzzy set defined on the set of values of variable V and Q is a Fuzzy number on R.

Any proposition P (given above) can be expressed as,

$$p': \text{There are } QE's$$

where E is a fuzzy set on a given I that is defined by,

$$E(i) = F(V(i)) \quad \forall i \in I \text{ and}$$

$$|E| = \sum_{i \in I} E(i) = \sum_{i \in I} F(V(i))$$

Also, the truth value is given by,

$$T(P) = T(P') = Q(|E|)$$

Example

P: There are about three students in I whose fluency in English, $V(i)$, is high

Let $I = \{\text{Bob, Adam, Eve, Cathy, David}\}$ and v is a Variable with values in the interval $[0,100]$ that express degrees of fluency in English.

In this case 'Q' is a fuzzy quantifier "about 3" shown in the figure (1). 'F' is a fuzzy set on $[0,100]$ and captures the linguistic term "high fluency" as shown in the figure (2).

Assume $V(\text{Bob}) = 20$, $V(\text{Adam}) = 35$, $V(\text{Eve}) = 70$, $V(\text{Cathy}) = 80$ and $V(\text{David}) = 95$.

The truth value of the proposition is,

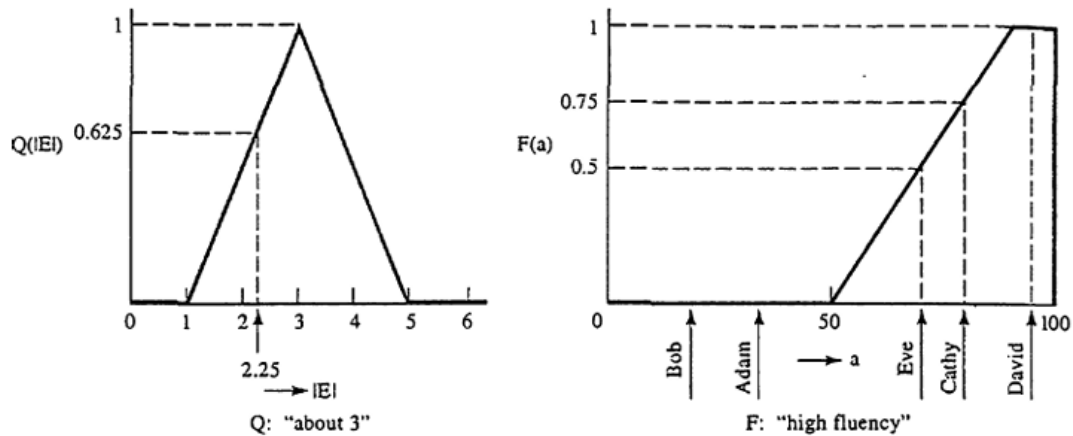


Figure 1.3: Fuzzy sets in a quantified fuzzy proposition

The fuzzy set is defined by

$\{0/\text{Adam}, 0/\text{Bob}, 0.75/\text{Cathy}, 1/\text{David}, 0.5/\text{Eve}\}$

$$|E| = \sum_{i \in I} E(i) = 0 + 0 + 0.75 + 1 + 0.5$$

$$= 2.25$$

$$T(P) = Q(|E|)$$

$$= Q(2.25)$$

$$= 0.625$$

The second form is,

P: There are Q 's in I such that $V_1(i)$ is F_1 and $V_2(i)$ is F_2 .

where V_1 and V_2 are variables.

The above proposition can be expressed in a simplified form,

P': QE_1 's E_2 's S _____(1)

where Q is the same quantifier as in the first kind and E_1, E_2 are fuzzy sets on I that are defined by the compositions,

$$E_1(i) = F_1(V_1(i))$$

$$E_2(i) = F_2(V_2(i))$$

For all $i \in I$, and eqn (1) can be interpreted as,

P': W is Q

where $W = |E_1 \cap E_2|$

$$= \sum_{i \in I} \min [F_1(V_1(i)), F_2(V_2(i))]$$

Then $T(P) = T(P) = Q(w)$.

2. Relative quantifiers

Relative quantifiers are quantifiers such as "about half", "most", "almost all", and so on. They are represented by fuzzy numbers on the unit interval $[0, 1]$.

The general form of the proposition is,

P: Among i 's in I such that $V_1(i)$ is F , there are Q i 's in I such that $V_2(i)$ is F_2 ,

where ' Q ' is a fuzzy number on $[0, 1]$.

The above proposition can be written as,

P': QE_1 's are E_2 's _____(1)

where E_1 and E_2 are fuzzy sets defined by

$$E_1(i) = F_1(V_1(i))$$

$$E_2(i) = F_2(V_2(i)) \quad \forall i \in I$$

Eqn(1) can be re-written as,

$$P': w \text{ is } Q$$

$$\begin{aligned} \text{where } W &= \frac{|E_1 \cap E_2|}{|E_1|} \\ &= \frac{\sum_{i \in I} \min[F_1(V_1(i)), F_2(V_2(i))]}{\sum_{i \in I} F_1(V_1(i))} \end{aligned}$$

$$\text{Then } T(P) = T(P') = Q(w)$$

Example

P: Among students in a given class that are young, there are at most all whose fluency in English is high.

4.5 Inference from Conditional and Qualified Propositions

The inference rule of our concern in this section involves conditional fuzzy propositions with fuzzy truth qualifiers. Given a conditional and qualified fuzzy proposition p of the form

$$p: \text{If } X \text{ is } A, \text{ then } y \text{ is } B \text{ is } S \text{ _____} (1)$$

where S is a fuzzy truth qualifier, and a fact is in the form " X is A' ," we want to make an inference in the form " y is B' ."

One method developed for this purpose, called a method of truth-value restrictions, is based on a manipulation of linguistic truth values. The method involves the following four steps.

Step 1. Calculate the relative fuzzy truth value of A' with respect to A , denoted by $RT(A'/A)$, which is a fuzzy set on the unit interval defined by

$$RT(A'/A)(a) = \sup_{x:A(x)=a} A'(x) \text{ _____} (2)$$

for all $a \in [0,1]$. The relative fuzzy truth value $RT(A'/A)$ expresses the degree to which the fuzzy proposition (8.46) is true given the available fact " X is A' ."

Step 2. Select a suitable fuzzy implication g by which the fuzzy proposition (1) is interpreted. This is similar to the selection of fuzzy implication, whose purpose is to express a conditional but unqualified fuzzy proposition as a fuzzy relation.

Step 3. Calculate the relative fuzzy truth value $RT(B'/B)$ by the formula

$$RT(B'/B)(b) = \sup_{a \in [0,1]} \min[RT(A'/A)(a), S(\partial(a, b))]$$

for all $b \in [0,1]$, where S is the fuzzy qualifier in (1). Clearly, the role of the qualifier S is to modify the truth value of $\partial(a, b)$. Note that when S stands for true (i.e., $S(a) = a$) for all $a \in [0,1]$, then $S(\partial(a, b)) = \partial(a, b)$.

The relative fuzzy truth value $RT(B'/B)$ expresses the degree to which the conclusion of the fuzzy proposition (1) is true.

Step 4. Calculate the set B' involved in the inference " y is B " by the equation

$$B'(y) = RT(B'/B)(B(y)). \text{---(3)}$$

for all $y \in Y$.

Example

Suppose we have a fuzzy conditional and qualified proposition,

p : If X is A then y is B is very true,

where $A = 1/x_1 + .5/x_2 + .7/x_3$, $B = .6/y_1 + 1/y_2$, and S stands for very true; let $S(a) = a^2$ for all $a \in [0,1]$. Given a fact " X is A' ," where $A' = .9/x_1 + .6/x_2 + .7/x_3$, we conclude that " y is B' ," where B' is calculated by the following steps.

Step 1. We calculate $RT(A'/A)$ by (2) :

$$\begin{aligned} RT(A'/A)(1) &= A'(x_1) = .9 \\ RT(A'/A)(.5) &= A'(x_2) = .6 \\ RT(A'/A)(.7) &= A'(x_3) = .7 \\ RT(A'/A)(a) &= 0 \text{ for all } a \in [0,1] - \{.5, .7, 1\}. \end{aligned}$$

Step 2. We select the Lukasiewicz fuzzy implication ∂ .

Step 3. We calculate $RT(B'/B)$

$$\begin{aligned} RT(B'/B)(b) &= \max\{\min[.9, S(\partial(.9, b))], \min[.6, S(\partial(.6, b))], \\ &\quad \min[.7, S(\partial(.7, b))]\} \\ &= \begin{cases} (.4 + b)^2 & \text{for } b \in [0, .375) \\ .6 & \text{for } b \in [.375, .475) \\ (.3 + b)^2 & \text{for } b \in [.475, .537) \\ .7 & \text{for } b \in [.537, .737) \\ (.1 + b)^2 & \text{for } b \in [.737, .849) \\ .9 & \text{for } b \in [.849, 1] \end{cases} \end{aligned}$$

A graph of this function $RT(B'/B)$ is shown in Fig.

Step 4. We calculate B' by (3):

$$\begin{aligned} B'(y_1) &= RT(B'/B)(B(y_1)) = RT(B'/B)(.6) = .7, \\ B'(y_2) &= RT(B'/B)(B(y_2)) = RT(B'/B)(1) = .9. \end{aligned}$$

Hence, we make the inference " y is B' ," where $B' = .7/y_1 + .9/y_2$.

When S in (1) stands for true (i.e., S is the identity function), the method of truth-value restrictions is equivalent to the generalized modus ponens under a particular condition, as stated in the following theorem.

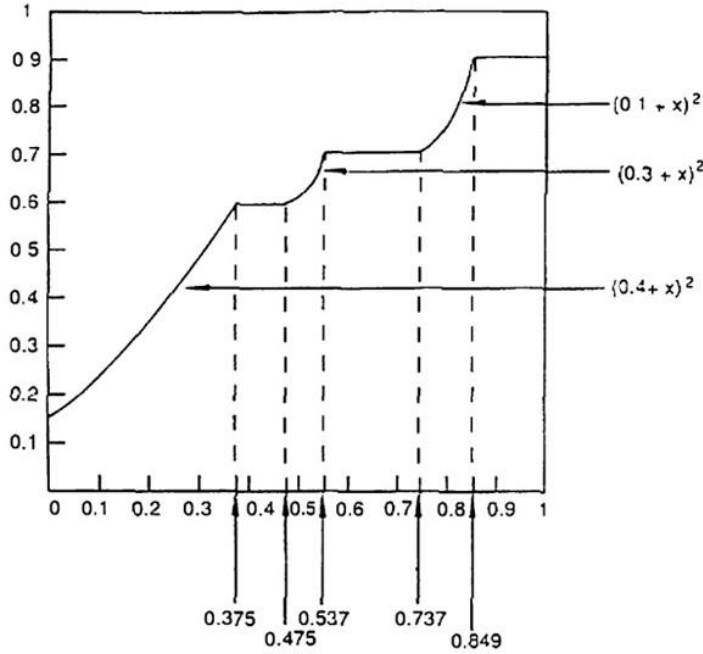


Figure 1.8: Function $RT(B/B')$ in Example 3.32.

Theorem. Let a fuzzy proposition of the form (1) be given, where S is the identity function (i.e., S stands for true), and let a fact be given in the form " X is A' ," where

$$\sup_{x:A(x)=a} A'(x) = A'(x_0) \text{---(4)}$$

for all $a \in [0,1]$ and some x_0 such that $A(x_0) = a$. Then, the inference " y is B' " obtained by the method of truth-value restrictions is equal to the one obtained by the generalized modus ponens define the same membership function B'), provided that we use the same fuzzy implication in both inference methods.

Proof: When $S(a) = a$ for all $a \in [0,1]$, B' , defined by (3), becomes

$$B'(y) = \sup_{a \in [0,1]} \min[RT(A'/A)(a), \partial(a, B(y))]\text{---(5)}$$

for all $y \in Y$. Using the same fuzzy implication J , B' becomes

$$B'(y) = \sup_{x \in X} \min[A'(x), \partial(A(x), B(y))]\text{---(6)}$$

for all $y \in Y$. To prove the theorem, we have to show that (8.51) and (8.52) define the same membership function B' . To facilitate the proof, let B'_1, B'_2 denote the functions defined by (5) and (6), respectively. Since

$$A'(x) \leq \sup_{x': A(x')=A(x)} A'(x') = RT(A'/A)(A(x))$$

for all $x \in X$, we have

$$\min[A'(x), \partial(A(x), B(y))] \leq \min[RT(A'/A)(A(x)), \partial(A(x), B(y))]$$

for all $y \in Y$. Hence,

$$\begin{aligned} B'_2(y) &= \sup_{x \in X} \min[A'(x), \partial(A(x), B(y))] \\ &\leq \sup_{x \in X} \min[RT(A'/A)(A(x)), \partial(A(x), B(y))] \\ &\leq \sup_{a \in [0,1]} \min[RT(A'/A)(a), \partial(a, B(y))] \\ &= B'_1(y) \end{aligned}$$

for all $y \in Y$. On the other hand, by condition (4), we have

$$\begin{aligned} \min[RT(A'/A)(a), \partial(a, B(y))] &= \min[\sup_{x: A(x)=a} A'(x), \partial(a, B(y))] \\ &= \min[A'(x_0), \partial(A(x_0), B(y))] \\ &\leq \sup_{x \in X} \min[A'(x), \partial(A(x), B(y))] \\ &= B'_2(y) \end{aligned}$$

for all $y \in Y$. Thus,

$$B'_1(y) = \sup_{a \in [0,1]} \min[RT(A'/A)(a), \partial(a, B(y))] \leq B'_2(y)$$

for all $y \in Y$ and, consequently, $B'_1 = B'_2$.

MODULE 5: APPLICATIONS OF FUZZY SET

Introduction

The concept of fuzzy set was developed in 1965. Since then, researchers have used this key set in many disciplines. As a brand-new conceptual system to support human-centric framework, fuzzy set has proved quite promising and effective in modelling human involvement in human-based intelligence to attain modernity in many departments like data analysing, data mining, image coding and explaining, as well as in intelligence systems. Fuzzy set has also become an acknowledged research subject in both pure as well as in applied mathematics and statistics, showing how this theory is highly applicable and productive in many applications. Despite being a core subject for many years, fuzzy set still attracts researchers for putting forth solutions for prime issues with certain features questioned by these notions. Fuzzy set can effectively deal with a wide spectrum of problems of the physical world via cooperation, which may be beyond the capability of classical techniques. This means the fuzzy set could have the ability to handle a wide range of problems, for instance, decision making, intelligent data analysis, processing information, pattern recognition, and optimization.

The goal of this Special Issue is to dive deeper into the new trends of fuzzy set theory and the extension of fuzzy set theory with applications in group theory, ring theory, statistics, topological spaces, graph theory, decision making and other engineering applications.

Economics

Economics, a social science addressing issues arising from relative scarcity, has evolved since the 19th century, with axiomatic theories progressing in precision and mathematical sophistication. Despite these advancements, there persists a gap between economic theories and reality. One reason for this gap, as suggested in the text, is the use of classical mathematics, which may not adequately capture the inherent uncertainty and complexity in economic phenomena.

Classical mathematics, rooted in classical set theory and two-valued logic, is considered unrealistic for economics. The text argues that human reasoning and decision-making involve genuine uncertainty, a concept not well-expressed in

classical mathematics. Additionally, human preferences for complex choices do not always adhere to the rules of additive measures.

Accurately modelling complex economic phenomena poses a challenge, and when complexity becomes overwhelming, simplification is necessary. However, the text notes that simplifying models for certainty and precision may result in a loss of relevance to the real world.

To address this issue, the text proposes introducing uncertainty into models, acknowledging that certainty and precision may be sacrificed under acceptable complexity limits. Fuzzy set theory emerges as a tool to express and deal with propositions in natural language, allowing for imprecise yet relevant economic predictions.

French economists, particularly Claude Ponsard, recognized the potential of fuzzy set theory in reformulating economic theory in the 1980s. There are two streams of activities in the reformulation of classical economic theory: one focusing on fuzzy set theory and the other on fuzzy measure theory, both aiming to make economic theory more realistic.

In microeconomics, the incorporation of fuzzy preferences seeks to reformulate traditional theories, introducing a more realistic depiction of economic decision-making. Fuzzy preferences accommodate uncertainty and subjectivity in individuals' choices, challenging assumptions prevalent in traditional economic models.

The exploration of equilibria in fuzzy noncooperative and cooperative games, as well as fuzzy economic equilibria, reveals that the introduction of fuzzy preferences has no impact on the existence of these equilibria. However, it suggests the possibility of unconventional equilibria in the presence of fuzzy preferences.

While the acceptance of efforts involving both fuzzy set theory and fuzzy measure theory has been lukewarm among economists, the acceptance of fuzzy measure theory seems to be growing more rapidly.

Civil Engineering

Civil engineering stands out among engineering disciplines due to the inherent uniqueness of each project, making available theories seldom fully applicable to actual design problems. The absence of prototype testing, common in other engineering fields, amplifies the uncertainty in applying theoretical solutions to

civil engineering projects. Despite the scientific advancements, civil engineering designers face the challenge of making decisions amidst high uncertainty, given the stringent safety standards associated with constructions like bridges, buildings, and dams.

Critical to a civil engineer's knowledge is an understanding of materials—how they are manufactured, shaped, assembled, and their resistance to stress, weather, and wear. Historically, civil engineering relied on subjective judgments and accumulated design experiences, even employing "rules of thumb" in the absence of scientific knowledge. Despite advancements, the uncertainty persists due to the uniqueness of each civil engineering design problem, necessitating the use of subjective judgments and relevant rules of thumb. This unique characteristic has led civil engineers to find affinity with fuzzy set theory.

Fuzzy set theory has proven useful in addressing problems related to assessing or evaluating existing constructions in civil engineering. Examples include assessing fatigue in metal structures, evaluating the quality of highway pavements, and determining damage in buildings post-earthquake. The text illustrates the application of fuzzy set theory through the assessment of the physical condition of bridges. With many aging bridges designed for lighter loads than current standards require, assessing their conditions is crucial for prioritizing rehabilitation or replacement.

The Bridge Inspector's Training Manual provides guidelines for bridge inspectors, outlining the assessment of bridge components, such as decks, superstructures, and substructures. Since inspections rely heavily on imprecise information and subjective judgment, fuzzy set theory becomes valuable. It allows expressing the condition and structural importance of each component as fuzzy sets, considering human subjectivity and imprecision. The overall evaluation of a bridge involves calculating the weighted average for all components using fuzzy arithmetic. The formula for the fuzzy set expressing the overall evaluation considers the condition and weight of each component.

In a simplified example with three components—deck, beams, and piers—evaluated linguistically as poor, fair, and good, fuzzy sets representing these conditions are applied. These fuzzy sets express the meaning of linguistic terms based on assumed condition ratings, allowing for a more nuanced assessment considering the imprecise nature of bridge inspections.

Mechanical Engineering

The application of fuzzy sets in engineering design, particularly in mechanical engineering, addresses the challenge posed by the importance of early design decisions. Recognizing that precise early decisions can be impractical and costly if later alternatives are restricted, experienced designers initiate the process with an imprecise description of the desired artifact. Fuzzy set theory proves valuable in these early stages, allowing designers to describe the artifact approximately and then refine the description as the design progresses.

In the context of mechanical engineering design, the basic concept is that fuzzy sets enable the representation of designed artifacts in approximate terms during the early stages. This approximation, described by imprecise input parameters, is used to calculate corresponding approximate characterizations of relevant output parameters. These characterizations are then compared with predefined performance criteria, and the information obtained guides the determination of appropriate values for input parameters.

The terminology used includes input parameters (design parameters), output parameters, and performance parameters. Input parameters are independent variables determined during the design process, output parameters depend on input parameters and possibly some performance parameters, and performance parameters have specified functional requirements. The expression of preferences for input parameters is done through appropriate fuzzy sets, reflecting the designer's desire for various parameter values. Triangular fuzzy numbers, representing linguistic descriptions, are mentioned as a convenient choice for expressing preferences in subjective terms.

In summary, fuzzy set theory is introduced as a valuable tool in the early stages of engineering design, allowing for imprecise descriptions and facilitating the decision-making process by incorporating subjective preferences and relevant engineering data. The focus is on mechanical engineering design, emphasizing the importance of addressing uncertainty in early design decisions.

Industrial Engineering

Fuzzy set theory proves valuable in this context by providing a flexible framework for dealing with imprecise information and uncertainties inherent in industrial systems.

One application of fuzzy sets in industrial engineering is in estimating the service life of equipment under various operating conditions. The challenge lies in obtaining a realistic estimate, often expressed in natural language. Fuzzy sets are used to represent normal operating conditions, and a fuzzy number is employed to express the estimated service life. The relationship between these sets is defined to account for the degree of nonspecificity in the estimate. This approach allows for a more comprehensive representation of the available information regarding the estimated service life and facilitates the estimation of service life under conditions different from normal.

Another application involves the design of built-in tests for industrial systems, particularly focusing on range tests. In the context of embedded control or monitoring systems, where computers play a crucial role, fuzzy set theory aids in evaluating the acceptability of quantities observed through a digital sampler. The imprecise nature of sampled data, limited by the resolution of the sampler, is addressed using fuzzy sets. This allows the computer to make educated guesses about the acceptability of quantities concerning specified operating ranges, enabling appropriate actions to be taken in response to detected anomalies.

In summary, industrial engineering, with its broad scope covering human, machine, material, and financial components, benefits from fuzzy set theory in addressing the complex, cross-disciplinary nature of industrial systems. Fuzzy control and fuzzy decision making are highlighted as well-developed applied areas within industrial engineering, with various other applications explored to varying extents. Fuzzy set theory provides a valuable tool for handling imprecise information, uncertainty, and behavioural aspects in the design, operation, and control of industrial systems.

Computer Engineering

Since the mid-1980s, the field of computer engineering has witnessed the emergence of fuzzy controllers and the subsequent demand for specialized hardware to implement fuzzy logic and approximate reasoning operations. The development of computer hardware tailored for fuzzy computing gained momentum in the late 1980s and reached fruition in the 1990s, resulting in commercially available hardware designed specifically for certain types of fuzzy computing tasks. This technological evolution primarily focuses on implementing fuzzy rules of inference and defuzzification procedures crucial for fuzzy controllers in various applications.

The primary motivation for utilizing specialized hardware in fuzzy computing within computer engineering lies in the quest for increased operational speed through parallel processing. Fuzzy computer hardware, whether in digital or analog mode, enables the parallel processing of all inference rules in a complex fuzzy inference engine. This enhanced efficiency significantly broadens the scope of applicability for fuzzy controllers and, potentially, other fuzzy expert systems.

In the digital mode of fuzzy computer hardware, fuzzy sets are represented as vectors of numbers in the $[0, 1]$ range. Operations on these fuzzy sets are conducted as vector operations, allowing for flexible programmability and seamless integration with existing predominantly digital computer systems. Conversely, the analog mode represents fuzzy sets using continuous electric signals, such as electric currents or voltages, through specific electric circuits. These circuits play a role akin to function generators in classical analog computers, offering high speed and compatibility with sensors.

Both modes of fuzzy computer hardware have distinctive advantages. Digital fuzzy hardware provides flexible programmability and seamless integration with existing digital systems, making it suitable for implementing complex schemes of multistage fuzzy inference. On the other hand, analog fuzzy hardware boasts high speed and excellent compatibility with sensors, making it particularly well-suited for the implementation of complex on-line fuzzy controllers.

In summary, the application of fuzzy set theory in computer engineering, particularly through the development of specialized fuzzy computer hardware, demonstrates a powerful tool for enhancing the operational speed and efficiency of fuzzy controllers. This advancement opens up new possibilities for applying fuzzy logic in diverse fields, contributing to the broader integration of fuzzy systems in computer engineering applications.

Robotics

Robotics is a cross-disciplinary subject area that is concerned with the design and construction of machines that are capable of human-like behavior. Such machines are referred to as robots. Although industries began interested in robots in the 1960s, the first commercial robots were produced only in the 1970s. These early robots, which are called industrial robots, were designed for repetitive actions of a typical production line. Advances in computer technology and in the area of artificial intelligence in the 1980s stimulated greater

ambitions in robotics. Research in robotics became oriented primarily on robots with high intelligence.

Intelligent robots must be equipped with at least three types of functions: appropriate perceptual functions facilitated by visual, auditory, and other sensors to allow the robot to perceive its environment; intelligent information processing functions that allow the robot to process the incoming information with respect to a given task; and mechanical functions with appropriate controls to allow the robot to move and act as desired.

To achieve human-like behaviour, an intelligent robot must have some key capabilities that distinguish human beings from classical nonintelligent machines. They include the capability of common-sense reasoning in natural language that is based on incoming information as well as background knowledge, high-level capabilities of pattern recognition and image understanding, refined motor controls, and the capability of making decisions in the face of bounded uncertainty. It is clear that the use of fuzzy set theory is essential for all these capabilities.

The main subjects of fuzzy set theory that are relevant to robotics include approximate reasoning, fuzzy controllers and other kinds of fuzzy systems, fuzzy pattern recognition and image processing, fuzzy data bases, information retrieval systems, and fuzzy decision making. In robotics, all these subjects, and a host of others, are utilized as components integrated in the overall architecture of intelligent robots. To cover this important role of fuzzy set theory in the formidable task of designing intelligent robots would require the introduction of a prohibitively extensive background information.

Medicine

The application of fuzzy set theory in the field of medicine has been recognized since the mid-1970s, primarily focusing on the uncertainty inherent in the process of diagnosing diseases. In this context, the increasing volume of information available to physicians from advanced medical technologies poses challenges in classifying symptom sets and determining appropriate therapeutic actions.

The complexity arises from the variability in how diseases manifest, with a single disease exhibiting diverse presentations in different patients and at various stages. Moreover, a single symptom may indicate multiple diseases, and the coexistence of several diseases in one patient can disrupt typical symptom patterns.

Medical descriptions often involve linguistic terms, introducing inherent vagueness. For example, statements about diseases like hepatitis may include terms such as "usually normal," "slightly decreased," or "increased," adding imprecision to the diagnostic process.

Physicians gather information about patients from various sources, including past history, physical examinations, laboratory tests, and imaging procedures. However, this information is fraught with uncertainty – past history may be subjective or incomplete, physical examinations can have mistakes, and measurements from tests may lack precision.

Despite these uncertainties, physicians must determine a diagnostic label to guide the appropriate therapeutic regimen. This emphasizes the critical nature of accurate and timely diagnosis.

Fuzzy set theory has been applied in various models to capture the complexity of the diagnostic process in medicine. These models consider factors such as the relative importance of symptoms, varied symptom patterns at different disease stages, relations between diseases, and the different stages within the diagnostic process.

Computerized medical expert systems, built on fuzzy set-based models, aim to assist physicians in diagnosing specific categories of diseases. These systems provide a framework to handle imprecise and uncertain information, contributing to more effective medical diagnosis.

In the realm of fuzzy set models for medical diagnosis, the physician's medical knowledge is often represented as a fuzzy relation between symptoms and diseases. The compositional rule of inference is employed to infer possible diseases based on observed symptoms, considering the uncertainty associated with both symptom manifestation and medical knowledge.

These applications of fuzzy set theory offer valuable tools for improving the understanding, teaching, and implementation of the complex process of medical diagnosis, providing a means to navigate and address uncertainties in the medical field.

Fuzzy Data Mining

With the development of electronic data processing, more and more data were available electronically. This led to the situation in which the masses of data, for

instance in data warehouses, exceeded the human capabilities to recognize important structures in these data.

Classical methods to 'data mine', such as cluster techniques, were available, but often they did not match the needs. Cluster techniques, for instance, assumed that data could be subdivided crisply into clusters, which did not fit the structures that existed in reality. Fuzzy set theory seemed to offer good opportunities to improve existing concepts.

Bezdek was one of the first who developed fuzzy cluster methods with the goals to:

- Search for structure in data
- Reduce complexity
- Provide input for control and decision making

Different approaches have been followed:

- Hierarchical approaches
- Semi-formal heuristic approaches
- Objective function clustering

The field of data mining is becoming increasingly important. Fuzzy methods have shown promise in data mining applications. Further development of fuzzy methods for data mining is expected.

CONCLUSION

The exploration of "FUZZY SET & FUZZY LOGIC, AND ITS APPLICATIONS" has provided valuable insights into the realm of fuzzy systems. Through the thorough examination of fuzzy operations, relations, and logic, as well as their practical applications, this project has underscored the versatility and efficacy of fuzzy theory in addressing real-world complexities. By delving into the foundational concepts and methodologies, as well as showcasing their diverse applications across various domains, this report serves as a comprehensive resource for understanding and leveraging fuzzy sets and fuzzy logic. As we continue to navigate increasingly complex and uncertain environments, the knowledge and insights gained from this project are invaluable for advancing the field and fostering innovation in problem-solving approaches.

Furthermore, this project underscores the significance of fuzzy systems in addressing ambiguity and uncertainty inherent in many real-world scenarios. By harnessing the power of fuzzy sets and fuzzy logic, we can effectively model and analyse complex systems where precise mathematical models may be inadequate.

The applications explored in this report, ranging from engineering and robotics to healthcare and decision-making processes, highlight the wide-ranging impact and relevance of fuzzy theory across diverse domains. As technology continues to evolve and our understanding of complex systems deepens, the principles and methodologies elucidated in this project provide a solid foundation for future research and innovation in the field of fuzzy logic and its applications.

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